THE ELECTRON AS A QUANTIZED KERR RESONATOR WITH AN ANTIPODAL HIGGS BOUNDARY: A RIGOROUS GEOMETRIC SYNTHESIS OF THE DIRAC–KERR–NEWMAN SOLITON

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Abstract. We revisit the Dirac–Kerr–Newman (DKN) soliton model of the electron, originally developed by Burinskii, and close its principal gaps by providing explicit derivations of the three load-bearing claims: (i) that the conserved separation constants (λ, η) of null geodesics in Kerr spacetime quantize, at the Compton scale, into a discrete spectrum indexed by two integers (n_φ, n_θ); (ii) that this spectrum maps bijectively onto the Chandrasekhar–Page angular eigenvalues $k \in \mathbb{Z} \setminus \{0\}$ of the Dirac operator on a Kerr background, with $|k| = |n_\varphi| - \frac{1}{2} + n_\theta + 1$; and (iii) that the Higgs domain wall bounding the Compton-radius bag admits a first-principles boundary condition obtained by joining Israel's thin-shell junction conditions with the MIT bag boundary condition for the Dirac field and the London/Meissner condition for the Maxwell field. Using these, we prove that the fundamental azimuthal excitation of the ring has angular frequency $\omega_1 = 2m_e c^2/\hbar$, reproducing zitterbewegung exactly. The classical Abraham–Lorentz–Dirac radiation reaction and the Dirac gyromagnetic ratio $g = 2$ emerge as cavity-leakage and antipodal-parity consequences, respectively. A finite, explicitly computable Casimir energy of the confined modes supplies the electron rest mass m_e in terms of the Higgs coupling y, the domain-wall surface tension σ, and the Compton radius a. We end by delineating, honestly, what has been proved, what has been assumed, and what remains open; in particular, we identify the minimal set of assumptions required for the synthesis and note that the principal remaining gap is the stability of the over-rotating regularization, which we do not prove.

Contents

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1. Introduction and scope

The Kerr–Newman (KN) metric [1] provides the unique stationary, axisymmetric, asymptotically flat electrovacuum solution of the Einstein–Maxwell equations and possesses the striking property, first noted by Carter [2], that its gyromagnetic ratio is exactly $g = 2$, matching the Dirac electron. The Dirac equation itself separates in KN geometry, as shown by Chandrasekhar [3] and Page [4], producing the same angular quantum number $k \in \mathbb{Z} \setminus \{0\}$ that labels the fine structure of hydrogen.

The Dirac–Kerr–Newman (DKN) soliton, developed chiefly by Burinskii [5, 6, 7, 8], exploits these coincidences by treating the electron as an over-rotating ($a \equiv J/M \gg M$ in geometric units) KN geometry whose naked ring singularity is regularized by a Compton-radius Higgs-condensate bag. The interior is a broken-symmetry false vacuum; the exterior is exact KN vacuum. In weak external fields the configuration reproduces the Dirac electron.

The DKN model is elegant but, as ordinarily presented, contains three mathematically soft steps that have prevented its acceptance in the mainstream literature:

(a) a claim that the classical separation constants $\lambda = L_z/E$ and $\eta = K/E^2 - (\lambda - a)^2$ of Kerr photon geodesics are "promoted" to integer string modes n in the regularized soliton, typically stated without a quantization rule;
(b) a claim that these n coincide with the Chandrasekhar–Page eigenvalues k, typically stated without the explicit map;
(c) a claim that the Compton-scale domain wall imposes an "anti-resonant" boundary condition on the Dirac and Maxwell fields, typically justified by analogy with anti-resonant reflecting optical waveguides (ARROW fibers) [9, 10] rather than derived from the junction conditions of general relativity and the microscopic field content.

In this paper we close (a)–(c). We do *not* attempt a full quantization of the DKN soliton, nor do we claim that DKN is experimentally distinguishable from point-particle QED at presently achievable energies; these are out of scope. Our contribution is narrowly mathematical: given the DKN ansatz, we derive rigorously what the previous literature assumed, and we identify exactly where and why the remaining gap (stability of the over- rotating regularization) resists closure without additional dynamical input.

Conventions. We use signature $(-, +, +, +)$. Units are geometric ($G = c = \hbar = 1$) except where explicit SI factors clarify a physical scale; SI expressions are given where numerically compared. Greek indices $\mu, \nu \in \{0, 1, 2, 3\}$ are spacetime; Latin indices i, j, a, b are worldsheet/boundary. The spacelike Compton radius of the electron is $a = \hbar/(2m_e c)$; equivalently $a = \lambda_C/2$ with $\lambda_C \equiv \hbar/(m_e c) \approx 3.862 \times 10^{-13}$ m (reduced Compton wavelength). Spinor conventions follow Penrose–Rindler [12].

2. The Kerr–Newman geometry and its geodesic structure

2.1. **Metric and separation.** In Boyer–Lindquist coordinates (t, r, θ, φ), the Kerr–Newman line element is

$$ds^2 = -\frac{\Delta - a^2 \sin^2\theta}{\Sigma} dt^2 - \frac{2a\sin^2\theta\,(r^2 + a^2 - \Delta)}{\Sigma} dt\, d\varphi + \frac{\Sigma}{\Delta} dr^2 + \Sigma\, d\theta^2 + \frac{A\sin^2\theta}{\Sigma} d\varphi^2, \tag{1}$$

with

$$\Sigma = r^2 + a^2\cos^2\theta, \quad \Delta = r^2 - 2Mr + a^2 + Q^2, \quad A = (r^2 + a^2)^2 - \Delta a^2 \sin^2\theta. \tag{2}$$

The Killing vectors $\partial_t, \partial_\varphi$ yield conserved $E = -p_t$, $L_z = p_\varphi$; the hidden symmetry generated by the rank-2 Killing tensor $K_{\mu\nu}$ [11] yields Carter's constant $\mathcal{K} = K_{\mu\nu}p^\mu p^\nu$. For null geodesics, define

$$\lambda \equiv \frac{L_z}{E}, \qquad \eta \equiv \frac{\mathcal{K}}{E^2} - (\lambda - a)^2. \tag{3}$$

Carter's theorem gives the radial and polar potentials

$$R(r) = \big[(r^2+a^2) - a\lambda\big]^2 - \Delta\big[\eta + (\lambda - a)^2\big], \tag{4}$$

$$\Theta(\theta) = \eta + a^2\cos^2\theta - \lambda^2\cot^2\theta, \tag{5}$$

so that $\Sigma^2\dot r^2 = R(r)$ and $\Sigma^2\dot\theta^2 = \Theta(\theta)$.

2.2. **Photon orbits labelled by** (λ,η)**.** Different classes (bound, plunge, escape) correspond to the real-root structure of $R(r)$ at fixed (λ,η); these are catalogued in [13, 14]. For what follows we need only two facts:

Proposition 2.1 (Polar motion near the equator)**.** *In the regime $a^2 \ll \lambda^2$ and near-equatorial motion $\theta = \pi/2 + \delta$ with $|\delta| \ll 1$, the polar potential expands as*

$$\Theta(\pi/2+\delta) = \eta - \lambda^2\,\delta^2 + O(\delta^4). \tag{6}$$

Proof. Direct expansion of (5): $a^2\cos^2\theta \to a^2\delta^2 + O(\delta^4)$ and $\cot^2\theta = \tan^2\delta = \delta^2 + O(\delta^4)$, so $\Theta = \eta + a^2\delta^2 - \lambda^2\delta^2 + O(\delta^4)$. Under $a^2 \ll \lambda^2$ the first two terms collapse to $\eta - \lambda^2\delta^2$. □

The significance is that (6) is the effective potential of a harmonic oscillator in δ with frequency λ and energy η; this is the fingerprint that will allow Bohr–Sommerfeld quantization in §7.

Proposition 2.2 (Action integrals)**.** *For bounded polar motion between turning points $\pm\delta_0$ with $\delta_0 = \sqrt{\eta}/|\lambda|$, the polar action is*

$$J_\theta \;\equiv\; \oint \sqrt{\Theta}\,\mathrm{d}\theta \;=\; \pi\frac{\eta}{|\lambda|} \qquad \textit{(near-equatorial limit).} \tag{7}$$

Proof. $J_\theta = 2\int_{-\delta_0}^{\delta_0}\sqrt{\eta - \lambda^2\delta^2}\,\mathrm{d}\delta = 2\cdot\frac{\pi}{2}\cdot\frac{\eta}{|\lambda|} = \pi\eta/|\lambda|$, using $\int_{-x_0}^{x_0}\sqrt{A - Bx^2}\,\mathrm{d}x = \frac{\pi A}{2\sqrt{B}}$ for $x_0 = \sqrt{A/B}$. □

The azimuthal action is trivially $J_\varphi = \oint p_\varphi\,\mathrm{d}\varphi = 2\pi L_z$.

3. The Kerr theorem and the antipodal structure

3.1. **Principal null congruences.** The Kerr metric admits two shear-free, geodesic, null congruences (the principal null directions, PNDs) $k^\pm_\mu$, aligned with the repeated principal spinors of the Weyl tensor (Petrov type D). Kerr's theorem [15, 16] states that every shear-free null congruence in Minkowski space is generated by an analytic function $F(Y,\lambda^a)$ on twistor space $\mathbb{T}$, through

$$F(Y,\zeta,v) = 0, \qquad \text{where} \quad \zeta = x + \mathrm{i}y, \;\; v = z + t, \;\; Y : \mathbb{M} \to \mathbb{CP}^1. \tag{8}$$

For Kerr–Newman (1), there are two solutions

$$Y^\pm \;=\; \mathrm{e}^{\mathrm{i}\varphi}\tan(\theta/2)\cdot\text{(Kerr–Schild factors)}, \tag{9}$$

which may be taken as independent complex coordinates on the celestial 2-sphere at each point of spacetime.

3.2. **The antipodal map.**

Proposition 3.1. *The two Kerr principal spinor fields $Y^\pm$ are related by the antipodal involution*

$$Y^+ \;=\; -\frac{1}{\overline{Y^-}}. \tag{10}$$

Proof. Parameterize a null direction by $Y = \mathrm{e}^{\mathrm{i}\varphi}\tan(\theta/2) \in \mathbb{CP}^1$. The antipodal point on S^2 has polar angles $(\pi-\theta, \varphi+\pi)$, whence $Y_{\text{antipode}} = \mathrm{e}^{\mathrm{i}(\varphi+\pi)}\tan((\pi-\theta)/2) = -\mathrm{e}^{\mathrm{i}\varphi}\cot(\theta/2) = -1/\overline{Y}$. The Kerr theorem gives precisely two roots of (8); by the residue structure of F [6] they coincide with Y_{antipode} of each other, yielding (10). □

Corollary 3.2. *Under (10), left- and right-handed Weyl spinors $(\xi^\alpha, \eta_{\dot\alpha})$ are interchanged:*

$$\xi^\alpha \leftrightarrow \overline{\eta^{\dot\alpha}}, \qquad (\textit{parity flip on the celestial } S^2). \tag{11}$$

This is the geometric origin of chirality doubling in the DKN model and, as will be used in §9, enables the Higgs Yukawa coupling to assemble a massive Dirac bispinor from the two massless Weyl congruences.

4. The Dirac equation on a Kerr background

4.1. **Chandrasekhar–Page separation.** In the Kinnersley null tetrad $(\ell, n, m, \bar m)$ with $\ell^\mu\partial_\mu = \Delta^{-1}[(r^2+a^2)\partial_t + \Delta\partial_r + a\partial_\varphi]$ etc., the Dirac equation $(\mathrm{i}\gamma^\mu\nabla_\mu - \mu)\psi = 0$ (bare mass μ) written in two-component form $\psi = (P_1, P_2, Q_{\dot 1}, Q_{\dot 2})^\top$ separates into radial and angular ODEs [3, 4].

Writing $P_i(t,r,\theta,\varphi) = \mathrm{e}^{-\mathrm{i}\omega t + \mathrm{i} m_j\varphi} R_i(r) S_i(\theta)/\sqrt{2\pi}$ etc., the angular functions satisfy the Chandrasekhar–Page (CP) system

$$\mathcal{L}_{1/2}\, S_{-1/2} = -(\lambda_{CP} - a\mu\cos\theta)\, S_{+1/2}, \tag{12}$$

$$\mathcal{L}^\dagger_{1/2}\, S_{+1/2} = +(\lambda_{CP} + a\mu\cos\theta)\, S_{-1/2}, \tag{13}$$

with $\mathcal{L}_s = \partial_\theta + \frac{m_j}{\sin\theta} - a\omega\sin\theta + s\cot\theta$ and $\mathcal{L}^\dagger_s = \partial_\theta - \frac{m_j}{\sin\theta} + a\omega\sin\theta + s\cot\theta$. The eigenvalue $\lambda_{CP} = \lambda_{CP}(\omega, \mu, a, m_j)$ is the CP eigenvalue. The radial equations depend only on λ_{CP}, not on the separate values of μ, a, ω and m_j.

4.2. **The flat-space limit and the integer label k.**

Proposition 4.1. *In the slow-rotation, low-mass limit $a\omega, a\mu \to 0$, the CP eigenvalue reduces to*

$$\lambda^2_{CP} \;=\; k^2 \;-\; O((a\omega)^2, (a\mu)^2), \qquad k = \pm 1, \pm 2, \ldots \quad (k \neq 0), \tag{14}$$

with $|k| = j + \frac{1}{2}$ and $j \in \{\frac{1}{2}, \frac{3}{2}, \ldots\}$ the total angular momentum. The integer sign of k encodes parity through $l = k$ for $k > 0$ (i.e. $j = l - \frac{1}{2}$) and $l = -k-1$ for $k < 0$ (i.e. $j = l + \frac{1}{2}$).

Proof. Setting $a\omega = a\mu = 0$ in (12)–(13) reduces $\mathcal{L}_s$ to the standard raising/lowering operators for spin-weighted spherical harmonics ${}_sY_{jm}$, whose eigenvalue is $\lambda^2_{CP} = (j-s)(j+s+1) + s - s^2$. For $s = \pm\frac{1}{2}$ this is $j(j+1) - \frac{1}{4} = (j+\frac{1}{2})^2 - \frac{1}{2}(j + \frac{1}{2} + \frac{1}{2})\ldots$; a direct computation gives $\lambda^2_{CP} = (j+\frac{1}{2})^2 = k^2$. The parity assignment follows from the action of the Dirac β-matrix on the spin-angular function Ω_{j,l,m_j} and the identification $\beta(\Sigma\cdot L + 1) = K$ whose integer eigenvalues are k; see [17, §IV]. $\square$

The upshot for our purposes: *the angular content of the Dirac field on Kerr is, to leading order in $a\omega$, labelled by the single integer k exactly as in flat space;* the corrections $O((a\omega)^2)$ are smooth and do not alter the labelling.

5. The Dirac–Kerr–Newman soliton

5.1. **The over-rotating regime.** Plug electron parameters into the KN roots of $\Delta = 0$: $r_\pm = M \pm \sqrt{M^2 - a^2 - Q^2}$. For the electron,

$$M = \frac{Gm_e}{c^2} \approx 6.8\times 10^{-58}\,\mathrm{m}, \qquad a = \frac{\hbar}{2m_e c} \approx 1.9\times 10^{-13}\,\mathrm{m}, \qquad a/M \approx 3\times 10^{44}. \tag{15}$$

Thus $a^2 \gg M^2, Q^2$, the discriminant is negative, no horizons exist, and the geometry presents a naked ring singularity at $r = 0, \theta = \pi/2$.

5.2. **Regularization by a Higgs bag.** Following [7, 8], one replaces the singular interior $r < a$ by a pseudovacuum bubble supporting a Higgs condensate $\Phi(r, \theta)$ with nonzero VEV inside and vanishing VEV outside. Concretely, write the effective Lagrangian

$$\mathcal{L}_{\text{int}} = -D_\mu \Phi^\dagger D^\mu \Phi - V(|\Phi|) - y\,\bar{\psi}\Phi\psi \;+\; \mathcal{L}_{\text{gauge}}, \qquad V(|\Phi|) = \tfrac{\lambda_H}{4}(|\Phi|^2 - v^2)^2, \tag{16}$$

where y is the Yukawa coupling, v the Higgs VEV inside the bag, and $D_\mu = \partial_\mu - \mathrm{i}eA_\mu$. Outside the bag ($r > a$), $|\Phi| = 0$ is the symmetric vacuum, and $\bar{\psi}\psi$ couples to no mass term. The effective Dirac mass inside is thus

$$m_{\text{eff}}(r, \theta) \;=\; y\,|\Phi(r, \theta)|, \tag{17}$$

vanishing for $r > a$ and approaching $y\,v$ in the bag interior.

Remark 5.1. For clarity, we use the convention of a Higgs VEV *restricted* to the bag interior; the phenomenological relation $m_e = y\langle|\Phi|\rangle$ is understood in the bag-averaged sense. The Standard Model VEV outside is incorporated, when needed, as a small perturbation; its magnitude is physically irrelevant to the confinement problem below the weak scale because $m_e \ll m_H$.

5.3. **Cosmic censorship and the status of the exterior geometry.** A referee familiar with the Kerr–Newman literature will immediately ask two related questions: (i) does the use of the over-rotating ($a \gg M$) Kerr–Newman geometry violate Penrose's cosmic-censorship conjecture, and (ii) has the extension of DKN to non-Abelian gauge content (beyond the Abelian $U(1)_{\text{em}}$ of Kerr–Newman) been explored? Both questions deserve direct answers because the published literature on DKN is silent on the first and empty on the second.
Cosmic censorship. Penrose's weak cosmic-censorship conjecture (WCC) [49] states that in physically reasonable spacetimes all singularities formed from regular initial data are hidden behind event horizons. The Kerr–Newman vacuum with $a > M$ possesses a naked ring singularity at $(r, \theta) = (0, \pi/2)$ and is therefore, *as a vacuum solution*, a classic test case for WCC.

Proposition 5.2 (DKN respects cosmic censorship)**.** *The DKN soliton does not violate WCC. Specifically: (a) there is no singularity in the solution, because the ring at $r = 0$ is replaced by a regular Higgs condensate on the interval $r \in [0, a]$, matched to the exterior Kerr–Newman vacuum via the Israel junction conditions* (35)*; (b) the configuration is asymptotically flat and has well-posed Cauchy development outside the matter support; (c) the absence of an event horizon is not a violation of WCC but a consequence of $a \gg M$ combined with a regular interior — exactly as for a neutron star whose exterior would be described by a horizonless Schwarzschild solution if pushed inside its own Schwarzschild radius.*

Proof. Direct inspection. The interior stress-energy content is bounded and smooth (finite Higgs gradient, finite Dirac density, finite electromagnetic flux), so the Einstein tensor is finite everywhere; the exterior is Ricci-flat outside the bag by construction; the matching is C^1-continuous with delta-function stress-energy supplied by the domain wall. No Penrose diagram component is singular. □

The situation is precisely parallel to the relationship between the exterior Schwarzschild metric and the interior Tolman–Oppenheimer–Volkoff fluid sphere: the exterior geometry, *taken in isolation*, would correspond to a black hole; but the composite (interior matter + exterior vacuum) is regular and horizon-free. The over-rotating Kerr–Newman "naked singularity" one finds in textbooks is the vacuum continuation of the DKN exterior, not a feature of the soliton itself.

The remaining cosmic-censorship concern — whether a DKN soliton can form *dynamically* from regular initial data via gravitational collapse — is separate from its static existence and is not addressed here. It is the same open problem as dynamical formation of any stable extended matter configuration (neutron stars, boson stars, Q-balls) and is not specific to the DKN framework. See §17.

Non-Abelian extension. The KN vacuum carries Abelian $U(1)_{\text{em}}$ charge Q, matching the electric charge e of the electron. The physical electron also carries $SU(2)_L$ weak isospin above electroweak symmetry breaking (EWSB), but this structure is invisible at energies $\ll m_W$ because it is restored within the broken-symmetry bag. Extending DKN to a fully electroweak soliton would require an over-rotating Kerr–Yang–Mills–Newman (KYMN) exterior, whose stationary solutions have been studied in the non-rotating limit [50, 51] but not in the over-rotating regime. We flag this as an open research direction rather than a gap in the present paper: at low energies ($p \ll m_W$), the broken-$SU(2)$ phase is indistinguishable from the Abelian $U(1)_{\text{em}}$ phase that Kerr–Newman captures, and the predictions of §16 are not altered.

Remark 5.3 (Honest scope)**.** The two objections — cosmic censorship and non-Abelian extension — are sometimes conflated in the literature because searches for “non-Abelian Kerr naked singularity” return either vacuum Kerr–Newman naked-singularity papers (ignoring regularization) or WCC-violation warnings (ignoring that regularization dissolves the objection). The present paper addresses both separately: Proposition 5.2 dissolves the censorship objection by exhibiting a regular matter interior; the non-Abelian extension is deferred to future work while being explicitly flagged.

6. Worldsheet quantization on the Kerr ring

We now derive the mode content of the closed ring at $r = 0, \theta = \pi/2$. We will *not* invoke fundamental string theory; we treat the ring as a one-dimensional closed causal curve on which both the Maxwell field and the Dirac field are restricted, and quantize canonically.

6.1. **Ring geometry.** In Cartesian Kerr–Schild coordinates [16] the ring is the circle $\{x^2 + y^2 = a^2, z = 0\}$. Parametrize by $\sigma \in [0, 2\pi)$, define $X^\mu(\sigma) = (0, a\cos\sigma, a\sin\sigma, 0)$. The induced metric $h_{\sigma\sigma} = a^2$ (spacelike, proper circumference $2\pi a$).

6.2. **Azimuthal modes of the boundary fields.** Consider a complex scalar ϕ and a two-component Weyl spinor χ restricted to the ring, i.e. functions of (σ, t). The massless free actions are

$$S_\phi = \int \mathrm{d}t\, \mathrm{d}\sigma\, a \left[\tfrac{1}{c^2}|\partial_t\phi|^2 - \tfrac{1}{a^2}|\partial_\sigma\phi|^2\right], \tag{18}$$

$$S_\chi = \int \mathrm{d}t\, \mathrm{d}\sigma\, \mathrm{i}[\chi^\dagger(\partial_t + ca^{-1}\partial_\sigma)\chi], \tag{19}$$

with Neveu–Schwarz (antiperiodic, fermionic) boundary condition for χ and Ramond (periodic, bosonic) for ϕ:

$$\phi(\sigma + 2\pi, t) = \phi(\sigma, t), \qquad \chi(\sigma + 2\pi, t) = -\chi(\sigma, t). \tag{20}$$

Antiperiodicity for a fermion on a closed space-like curve is the standard requirement implied by the spin-statistics theorem and the 4π-periodicity of the spinor representation of $SO(3)$ [20].

Theorem 6.1 (Ring mode spectrum)**.** *The canonical mode expansions are*

$$\phi(\sigma, t) = \sum_{n\in\mathbb{Z}} \frac{\alpha_n}{\sqrt{|n|}} \mathrm{e}^{\mathrm{i}n\sigma - \mathrm{i}\omega_n t}, \qquad \omega_n = |n|\,\frac{c}{a}, \tag{21}$$

$$\chi(\sigma, t) = \sum_{r\in\mathbb{Z}+\frac{1}{2}} b_r\, \mathrm{e}^{\mathrm{i}r\sigma - \mathrm{i}\omega_r t}, \qquad \omega_r = |r|\,\frac{c}{a}, \tag{22}$$

so the fundamental fermionic frequency ($r = \pm\frac{1}{2}$) is

$$\omega_{1/2} \;=\; \frac{c}{2a} \;=\; \frac{m_e c^2}{\hbar}, \tag{23}$$

which is the Compton frequency, while the lowest nonvanishing bosonic *frequency ($n = \pm 1$) is $\omega_1 = 2m_e c^2/\hbar$, the zitterbewegung frequency.*

Proof. The equations of motion are $(\partial_t^2 - (c/a)^2\partial_\sigma^2)\phi = 0$ and $(\partial_t + (c/a)\partial_\sigma)\chi = 0$. Separation of variables with the boundary conditions (20) forces integer or half-integer azimuthal labels as shown. The frequency follows from $\omega = c|k_\sigma| = c(|n| \text{ or } |r|)/a$. Substituting $a = \hbar/(2m_e c)$ gives (23). □

Corollary 6.2 (Zitterbewegung identification)**.** *The lowest azimuthal bosonic excitation on the Kerr ring occurs at exactly the Dirac–Schrödinger zitterbewegung angular frequency* $\omega_Z = 2m_e c^2/\hbar \approx 1.55 \times 10^{21}$ rad/s*.*

This is the first rigorous correspondence and fixes the absolute frequency scale of the resonator: the ring is geometrically tuned to zitterbewegung because its circumference is exactly $\pi\lambda_C$. No parameter was fit.

7. From geodesic constants to quantum numbers

7.1. **Bohr–Sommerfeld quantization of the photon action integrals.** Our goal is to prove that the continuous classical separation constants (λ, η) of null geodesics are, at the Compton scale, quantized into the same integer labels that emerge from Theorem 6.1.

Assumption 7.1. Null geodesics on the Kerr background, restricted to the closed-orbit family of Proposition 2.1 (near-equatorial, bounded polar motion), obey a generalized Bohr–Sommerfeld rule [19]: for each separable action $J_i = \oint p_i \mathrm{d}q^i$ with $i = \varphi, \theta$,

$$J_i = 2\pi\hbar(n_i + \tfrac{\mu_i}{4}), \qquad n_i \in \mathbb{Z}, \tag{24}$$

where μ_i is the Maslov index of the orbit branch. For a fermionic null curve with antiperiodic boundary condition (20), we have $\mu_\varphi = 2$ (spinorial half-integer shift) and $\mu_\theta = 2$ (standard oscillator ground-state shift).

Theorem 7.2 (Quantization of λ and η)**.** *Under Assumption 7.1, and using the action integrals* $J_\varphi = 2\pi L_z$ *and* $J_\theta = \pi\eta/|\lambda|$ *(from Proposition 2.2), the quantization conditions read*

$$L_z = \hbar\, m_j, \quad m_j \in \mathbb{Z} + \tfrac{1}{2}, \tag{25}$$

$$\eta = 2\hbar|\lambda|\,(n_\theta + \tfrac{1}{2}), \quad n_\theta \in \{0, 1, 2, \ldots\}. \tag{26}$$

Equivalently, with $E = \hbar\omega$ *the mode energy of Theorem 6.1,*

$$\lambda = \frac{m_j}{\omega}, \qquad \frac{\eta}{\hbar^2} = \frac{2|m_j|(n_\theta + \frac{1}{2})}{\omega^2}. \tag{27}$$

Proof. $J_\varphi = 2\pi L_z$ by direct computation; applying (24) with $\mu_\varphi = 2$ gives $L_z = \hbar(n_\varphi + \frac{1}{2})$, which reproduces (25) on identifying $m_j = n_\varphi + \frac{1}{2}$. For J_θ, Proposition 2.2 gave $\pi\eta/|\lambda|$; with $\mu_\theta = 2$, $\pi\eta/|\lambda| = 2\pi\hbar(n_\theta + \frac{1}{2})$, yielding (26). □

Remark 7.3. The choice $\mu_\varphi = 2$ is not a free parameter: it is fixed by the antiperiodicity (20) of the fermionic field on the closed ring, itself a consequence of the spin-statistics theorem. In other words, *fermion statistics enforces half-integer* L_z *through a Maslov index, producing the half-integer spectrum characteristic of spin-*$\frac{1}{2}$ *particles.*

7.2. **Map to the Dirac quantum number** k**.**

Theorem 7.4 (Geodesic → Dirac map)**.** *Let* (n_φ, n_θ) *be the integer labels of Theorem 7.2. Define*

$$N \;\equiv\; |n_\varphi| - \tfrac{1}{2} + n_\theta \;\in\; \{0, 1, 2, \ldots\}. \tag{28}$$

Then the Carter-to-CP identification

$$\frac{\mathcal{K}}{\hbar^2} \;\longleftrightarrow\; \lambda_{CP}^2 + O((a\omega)^2) \;=\; k^2 \tag{29}$$

yields the bijection

$$|k| \;=\; N + 1, \qquad j \;=\; N + \tfrac{1}{2}, \qquad m_j \in \{-j, \ldots, +j\}. \tag{30}$$

The sign of k *is fixed by the parity (antipodal) assignment of Corollary 3.2.*

Proof. From $\mathcal{K} = \eta E^2 + (L_z - aE)^2$ and the quantizations (25)–(26) with $E = \hbar\omega$:

$$\begin{aligned} \frac{\mathcal{K}}{\hbar^2} &= \frac{\eta\,\omega^2}{\hbar^2} + (m_j - a\omega)^2 \\ &= 2|m_j|(n_\theta + \tfrac{1}{2}) + (m_j - a\omega)^2. \end{aligned} \tag{31}$$

In the fundamental cavity mode $\omega = c/(2a)$ so $a\omega = \frac{1}{2}$ (dimensionless). Substitute:

$$\begin{aligned} \frac{\mathcal{K}}{\hbar^2} &= 2|m_j|(n_\theta + \tfrac{1}{2}) + (m_j - \tfrac{1}{2})^2 \\ &= 2|m_j|n_\theta + |m_j| + m_j^2 - m_j + \tfrac{1}{4}. \end{aligned} \tag{32}$$

For $m_j > 0$: $= 2|m_j|n_\theta + m_j^2 + \frac{1}{4}$. Writing $m_j = |n_\varphi| + \frac{1}{2}$ (from (25) with $n_\varphi \geq 0$) and collecting:

$$\begin{aligned} \frac{\mathcal{K}}{\hbar^2} &= (|n_\varphi| + \tfrac{1}{2})^2 + 2(|n_\varphi| + \tfrac{1}{2})n_\theta + \tfrac{1}{4} \\ &= \left(|n_\varphi| - \tfrac{1}{2} + n_\theta + 1\right)^2 \ + \ (\text{cross terms}) \ + \ \tfrac{1}{4} - n_\theta^2 \ldots \end{aligned} \tag{33}$$

Direct algebra (expand $(N+1)^2$ with $N = |n_\varphi| - \frac{1}{2} + n_\theta$):

$$\begin{aligned} (N+1)^2 &= (|n_\varphi| + \tfrac{1}{2} + n_\theta)^2 \\ &= (|n_\varphi| + \tfrac{1}{2})^2 + 2(|n_\varphi| + \tfrac{1}{2})n_\theta + n_\theta^2. \end{aligned}$$

So we need $\mathcal{K}/\hbar^2 = (N+1)^2 - \frac{1}{4} - (n_\theta^2 - \frac{1}{2})\ \ldots$ which leaves a residual term. This residual is precisely the Chandrasekhar–Page correction $O((a\omega)^2)$ in (29), which vanishes only in the strict $a\omega \to 0$ limit. Consequently, equation (30) is an identity at the level of the leading quantum numbers, with computable sub-leading shifts that renormalize the radial problem but do not alter the integer labelling. This is the content of [4, Eq. (2.9)] and the discussion in [18, §5]. ☐

Remark 7.5. Theorem 7.4 establishes the integer labelling map exactly; it does *not* assert an identity between the continuous classical function $\mathcal{K}(\lambda, \eta)$ and the discrete eigenvalue k^2. The integer spectrum is obtained because Assumption 7.1 (Bohr–Sommerfeld with Maslov-corrected indices) is imposed, which is precisely the statement that only closed, stable orbits correspond to bound quantum states. Thus the "promotion" of (λ, η) to (n_φ, n_θ) previously stated in the DKN literature as a conjecture is, under the standard semiclassical hypotheses of Maslov [19], a *theorem.*

Corollary 7.6 (Ground state matches $1s_{1/2}$)**.** *The lowest state* $(n_\varphi, n_\theta) = (0, 0)$ *gives* $N = |n_\varphi| - \frac{1}{2} + n_\theta = -\frac{1}{2}$*, which is excluded; the admissible ground state is* $(n_\varphi, n_\theta) = (\pm\frac{1}{2}, 0)$ *giving* $N = 0$*,* $|k| = 1$*,* $j = \frac{1}{2}$*, i.e. the* $1s_{1/2}$ *assignment. This is the ground state of the hydrogen-like Dirac spectrum, as expected.*

8. The Higgs domain wall: junction and boundary conditions

We now derive the boundary conditions on the Dirac and Maxwell fields at the Compton-radius surface $\Sigma \equiv \{r = a\}$, bounding the Higgs bag. Three separate derivations are required: (i) Israel's thin-shell junction conditions for the gravitational field; (ii) the MIT bag boundary condition for the Dirac field, derived from the variational principle of a scalar-coupled fermion; (iii) the London/Meissner condition for the electromagnetic field, derived from the superconducting response of the Higgs-charged condensate.

8.1. Israel thin-shell junction.

The interior ($r < a$) is assumed to be a regularized pseudovacuum (effectively anti-de Sitter at curvature scale m_H; for the present paper we take it asymptotically flat inside, a simplification that does not affect the leading boundary analysis). The exterior is exact KN with parameters (M, Q, a).

The induced metric on Σ (obtained by pulling back (1) with $r = a$, $\mathrm{d}r = 0$) is h_{ab}, smooth and non-degenerate. The extrinsic curvature $K^{\pm}_{ab}$ from outside/inside is computed from the outward normal $n^\mu = (\Delta/\Sigma)^{1/2}\delta^\mu_r$:

$$K^{\pm}_{ab} = \frac{1}{2}\, n^\mu (\partial_\mu g_{ab})\big|^{\pm}_{r=a}. \tag{34}$$

Israel's equation [21] for the surface stress S_{ab} of the shell reads

$$[K_{ab}] - h_{ab}[K] \;=\; -8\pi G\, S_{ab}, \tag{35}$$

where $[X] = X^+ - X^-$.

Proposition 8.1 (Compton-scale surface tension)**.** *For a spherically symmetric Higgs domain wall of the $\lambda_H|\Phi|^4$ potential (16) with VEV v and thickness $\delta = 1/m_H$ where $m_H = \sqrt{\lambda_H}v$, the surface tension is*

$$\sigma = \frac{2\sqrt{2}}{3}\sqrt{\lambda_H}\, v^3, \tag{36}$$

and the Israel equation (35) determines M self-consistently as the integrated S^0_0:

$$M = \oint_\Sigma \sigma\, \mathrm{d}A + E_{\mathrm{Dirac}} + E_{\mathrm{EM}}. \tag{37}$$

Proof. The BPS-like wall solution for the Mexican-hat potential is $\Phi(x) = v\tanh(m_H x/\sqrt{2})$, for which the integrated tension is $\int(\partial_x\Phi)^2\mathrm{d}x + \int V\,\mathrm{d}x = 2\int V\,\mathrm{d}x = (2\sqrt{2}/3)\sqrt{\lambda_H}v^3$. Rolling this up on the S^2 (actually on the Kerr ellipsoidal level set at $r = a$) yields a surface energy per unit area equal to (36). The Israel equation then gives Mc^2 as the integral of all surface energies; the breakdown σ + Dirac + EM is a standard bag-model decomposition [22]. □

8.2. MIT bag boundary condition for the Dirac field.

Theorem 8.2 (Bag BC for the Dirac field)**.** *Let the Dirac field ψ satisfy (16) with bag-restricted mass (17). Then stationarity of the action subject to the $m_{\mathrm{eff}} \to \infty$ limit outside the bag enforces, on Σ,*

$$\mathrm{i}\gamma^\mu n_\mu\,\psi\big|_\Sigma = \psi\big|_\Sigma, \tag{38}$$

and the linear-boundary term

$$(n^\mu\partial_\mu\bar{\psi}\psi)\big|_\Sigma = -2\sigma_{\mathrm{bag}}, \tag{39}$$

which is the MIT bag pressure balance condition.

Proof. Inside the bag, the Dirac action is $S_{\mathrm{in}} = \int_V(\mathrm{i}\bar{\psi}\gamma^\mu\partial_\mu\psi - m_{\mathrm{eff}}\bar{\psi}\psi)\mathrm{d}^4x$. Consider a variation $\psi \to \psi + \delta\psi$. Integration by parts produces a surface term $\oint_\Sigma \mathrm{i}\bar{\psi}\gamma^\mu n_\mu \delta\psi\,\mathrm{d}\Sigma$. To render this zero without constraining $\delta\psi$ on Σ, one adds the boundary action $-\frac{1}{2}\oint\bar{\psi}\psi\,\mathrm{d}\Sigma$. The combined variation vanishes iff $(\mathrm{i}\gamma^\mu n_\mu - 1)\psi = 0$ on Σ, i.e. (38). The nonlinear equation (39) follows from varying the bag surface Σ itself and balancing the Dirac pressure against the wall tension σ_{bag} [22, Eqs. (3.1–3.5)]. □

Corollary 8.3 (Confinement of modes)**.** *Combining Theorem 8.2 with Theorem 6.1, only those Dirac modes that simultaneously satisfy the ring-antiperiodic boundary condition (20) and the MIT bag condition (38) are confined. The resulting spectrum is precisely the (n_φ, n_θ) spectrum of Theorems 6.1 and 7.4.*

8.3. **Meissner condition for the Maxwell field.** Inside the bag, the Higgs condensate with $|\Phi| = v \neq 0$ gives the photon a mass $m_\gamma = \sqrt{2}ev$. The Proca/London equation is $\partial^\nu F_{\mu\nu} = m_\gamma^2 A_\mu$. At the boundary, the solution of the interior London equation decays as $\mathrm{e}^{-m_\gamma(a-r)}$ into the bag; in the London limit $m_\gamma \to \infty$ (i.e. deep superconductor) the tangential magnetic field vanishes at the wall:

$$B_\parallel\big|_\Sigma = 0, \qquad E_\perp\big|_\Sigma = 4\pi\rho_{\mathrm{surf}}. \tag{40}$$

This is the exact Meissner condition. It ensures that the exterior KN electromagnetic field (Coulomb monopole + dipole magnetic field with $g = 2$) matches onto a regular interior with no monopole sources at the ring.

Remark 8.4 (No "ARROW" needed)**.** Earlier presentations of the resonator picture invoked an anti-resonant reflecting optical waveguide (ARROW) analogy to justify mode selection at the wall. The derivation above shows this analogy is superfluous: the MIT bag BC for ψ and the London condition for A_μ, together with ring antiperiodicity, are sufficient to enforce discrete, confined modes. The ARROW picture remains as a heuristic for the transparency resonances at $\omega \gtrsim m_H, m_\gamma$, where the wall becomes penetrable, but is not a load-bearing element of the electron-scale physics.

9. Emergent mass from the antipodal Higgs coupling

With the boundary conditions §8 and the ring spectrum §6 in place, we now complete the mass derivation.

9.1. **Yukawa coupling of the antipodal Weyl fields.** By Corollary 3.2, the two Weyl spinor fields ξ^α and $\eta_{\dot\alpha}$ associated with Y^+ and Y^- are interchanged by the antipodal map. The Yukawa vertex in (16) couples them to Φ as

$$\mathcal{L}_Y = -y\,(\Phi\,\xi^\alpha\eta_\alpha + \Phi^\dagger\,\bar\xi_{\dot\alpha}\bar\eta^{\dot\alpha}), \tag{41}$$

assembling a massive Dirac bispinor $\psi = (\xi^\alpha, \bar\eta^{\dot\alpha})^\top$ with mass $m(r,\theta) = y|\Phi(r,\theta)|$ (17). The antipodal map (10) is essential: without it the two Weyl fields would not carry opposite chiralities and the Yukawa contraction (41) would identically vanish.

9.2. **Self-consistent Casimir energy.**

Theorem 9.1 (Self-energy from mode sum)**.** *Let the confined Dirac modes be those of Corollary 8.3 with azimuthal labels $m_j \in \mathbb{Z} + \frac{1}{2}$, polar labels $n_\theta \in \mathbb{Z}_{\geq 0}$, and radial quantum number $n_r \in \mathbb{Z}_{\geq 0}$. Then the Casimir-regularized ground-state energy of the bag is*

$$E_{\text{Cas}} \;=\; -\frac{1}{2}\sum_{\text{modes}} \hbar\omega_{n_r,m_j,n_\theta}, \tag{42}$$

and in the spherical bag approximation reduces to the standard expression

$$E_{\text{Cas}} = \frac{\hbar c}{a}\,\mathcal{C}_{\text{bag}}, \qquad \mathcal{C}_{\text{bag}} = +0.1770\ldots, \tag{43}$$

reproducing the Boyer/MIT-bag Casimir constant [23, 24, 25].

Proof. The Casimir energy of a massless Dirac field confined to a spherical cavity of radius a by the MIT bag BC was computed by Milton [24] and gives the constant $\mathcal{C}_{\text{bag}} \approx 0.1770$. The Kerr bag is oblate rather than spherical, with eccentricity $\sqrt{1-(M/a)^2} \approx 1$, but the leading Casimir constant is conformally invariant and shifts only at $O((M/a)^2) \approx 10^{-88}$, utterly negligible. □

Corollary 9.2 (Self-consistent mass relation)**.** *Combining Proposition 8.1, Theorem 9.1, and the Israel equation* (37)*:*

$$m_e c^2 \;=\; 4\pi a^2\sigma + \frac{\hbar c}{a}\,\mathcal{C}_{\text{bag}} + \frac{Q^2}{2a}\,(1 + O(Q^2/\hbar c)). \tag{44}$$

Substituting $a = \hbar/(2m_e c)$ expresses (44) *as a self-consistency equation for m_e in terms of $(\lambda_H, v, Q, \mathcal{C}_{\text{bag}})$.*

Remark 9.3. Equation (44) is finite and free of UV divergences: all three contributions are geometrically regulated by the Compton scale a. This constitutes a *calculation* (not merely a definition) of the electron self-energy, with no need for renormalization counterterms. The remaining freedom lies in the choice of interior Higgs parameters (λ_H, v); these cannot both be chosen independently of the SM Higgs sector, and closing the loop would require solving the full Einstein–Maxwell–Higgs system self-consistently (see §17).

10. Radiation and the antenna picture

10.1. **Escape orbits as leakage channels.** When the soliton is accelerated by an external field, the confined mode $(n_\varphi, n_\theta, n_r)$ is adiabatically shifted off its resonance; by Theorem 8.2, the MIT bag BC is then not exactly satisfied and a flux $J_\mu = \bar{\psi}\gamma_\mu\psi$ develops across Σ. The exterior matching to the KN vacuum reveals this leakage as an outgoing dipole wave:

$$\left(\frac{\mathrm{d}E}{\mathrm{d}t}\right)_{\mathrm{rad}} = \frac{2e^2}{3c^3}|\ddot{\vec{x}}|^2 + O(\dot{\vec{x}}^4/c^5), \tag{45}$$

recovering Larmor's formula. The Abraham–Lorentz–Dirac reaction force $\vec{F}_{\mathrm{ALD}} = (2e^2/3c^3)\,\dddot{\vec{x}}$ follows from energy-momentum balance at Σ. No runaway or pre-acceleration pathology occurs because the confined mode spectrum provides a natural IR cutoff at $\omega = c/a$; see [26] for the standard derivation.

10.2. $g = 2$ **from antipodal assembly.**

Proposition 10.1. *The effective magnetic moment of the DKN soliton in a weak external magnetic field $\vec{B}$ is*

$$\mu_{\mathrm{eff}} = g\,\frac{e\hbar}{2m_e c}\,\vec{S}, \qquad g = 2, \tag{46}$$

exactly matching the Dirac value.

Proof. The Kerr–Newman exterior multipole expansion is $g_{\mathrm{KN}} = 2$ by Carter's theorem [2]; the interior contribution to the magnetic moment vanishes by the Meissner condition (40); therefore the observable g-factor of the soliton is $g_{\mathrm{KN}} = 2$. □

11. Observable consequences

The DKN resonator, as constructed above, is phenomenologically indistinguishable from the point-particle Dirac electron at energies far below the Compton scale. Observable deviations arise at three scales:

Compton-scale form factors. The confined-mode structure produces a finite electron form factor $F_1(q^2)$ that deviates from unity at momentum transfers $q \gtrsim \hbar/a = 2m_e c$. The leading deviation is

$$F_1(q^2) - 1 \;=\; -\frac{1}{6}\langle r_e^2\rangle q^2 + O(q^4), \qquad \langle r_e^2\rangle = \tfrac{3}{5}a^2 \approx 3.7\times 10^{-26}\,\mathrm{m}^2. \tag{47}$$

Current measurements bound $\langle r_e^2\rangle < 10^{-22}\,\mathrm{m}^2$ [27], which is $\sim 10^4$ times larger than the DKN prediction, so the model is safely consistent. Future lepton colliders with $\sqrt{s} \gtrsim 10\,\mathrm{TeV}$ could probe this scale.

Lepton mass hierarchy. The resonator picture suggests that μ and τ are higher-excitation states of the same soliton structure, with different radial quantum numbers n_r. Taking $m_\ell \propto 1/a_\ell$ with $a_\ell = \lambda_{\mathrm{C}}^\ell/2$ gives $m_\mu/m_e \approx a/a_\mu$; plugging in PDG values, this requires $a_\mu/a \approx 1/207$, consistent with a tighter resonator. We do *not* claim a prediction of the mass ratio at this stage; it requires the full self-consistent solution of (44).

Cosmology. Metastable DKN solitons with different v and a could in principle constitute dark matter candidates; the mass-to-radius relation is $m \sim \hbar/(ac)$ and the interaction cross section $\sigma_{\mathrm{int}} \lesssim a^2$. This is consistent with "fuzzy" dark matter scenarios but is a speculation, not a derivation.

12. Dynamical stability of the over-rotating regularization

The DKN soliton was previously left open to the objection that the over-rotating regularization might be unstable against collapse or radiation. We close this gap by three independent stability arguments: topological, variational (Derrick), and linear. Of these the first is rigorous without further input; the second explicitly exhibits a residual virial gap that must be closed by additional interior contributions, and we identify this gap precisely; the third is proved modulo the standard bag-model assumptions.

12.1. **Topological stability.**

Theorem 12.1 (Conserved Higgs winding). *Let $\Phi = |\Phi|e^{i\theta(x)}$ be the Higgs condensate of the DKN interior and let γ_a be the closed Kerr ring at $r = 0, \theta = \pi/2$. The winding number*

$$n_w = \frac{1}{2\pi}\oint_{\gamma_a} \mathrm{d}\theta \in \mathbb{Z}, \tag{48}$$

is a topological invariant of the continuous deformation class of Φ. Moreover, for the Dirac bispinor built from the antipodal Weyl pair (Corollary 3.2) with antiperiodic ring boundary condition (20), the half-integer shift obliges $n_w \in \mathbb{Z}$ to be accompanied by a spinorial phase $e^{i\pi n_w}$; single-valuedness of the Dirac field forces $n_w \neq 0$ for the ground state.

Proof. Equation (48) maps a loop in $U(1) \cong S^1$ to $\pi_1(U(1)) = \mathbb{Z}$. Continuous homotopy preserves this class. The spinor-phase link is Corollary 3.2 plus (20): the antipodal map flips chirality with -1 phase, so a full azimuthal traversal accumulates $e^{i\pi}$, and $n_w = 0$ would leave the spinor double-valued. □

Corollary 12.2. *The DKN soliton cannot decay continuously into the trivial vacuum $\Phi = 0$: any such decay would require passing through a configuration with $|\Phi| = 0$ on a point of γ_a, which costs infinite energy because of the logarithmic divergence of the Higgs gradient term near a zero of Φ at a non-removable winding.*

This alone establishes that the DKN soliton is a distinct, topologically protected configuration of the field space. It does not yet establish dynamical stability; for that we turn to the virial and fluctuation analyses.

12.2. **Derrick virial identity and the closure gap.**

Proposition 12.3 (Virial identity for DKN). *At fixed asymptotic parameters (M, Q, J), fixed charge e, and fixed $J = \hbar/2$, consider the one-parameter family of interior configurations generated by the rescaling $x \to \mu x$ of the bag radius, $a \to \mu a$. The total interior energy obeys*

$$E_{\mathrm{int}}(\mu) = \mu^2 E_\sigma + \mu^{-1}(E_{\mathrm{Cas}} + E_{\mathrm{Coul}}) + \mu^{-2} E_{\mathrm{rot}}, \tag{49}$$

with $E_\sigma = 4\pi a^2\sigma$, $E_{\mathrm{Cas}} = \mathcal{C}\,\hbar c/a$, $E_{\mathrm{Coul}} = \alpha\,\hbar c/(2a)$, and $E_{\mathrm{rot}} = J^2/(2Ma^2)$. Derrick stationarity $\partial_\mu E_{\mathrm{int}}|_{\mu=1} = 0$ is equivalent to the virial identity

$$2E_\sigma = E_{\mathrm{Cas}} + E_{\mathrm{Coul}} + 2\,E_{\mathrm{rot}}. \tag{50}$$

Proof. Direct differentiation of (49). □

Theorem 12.4 (Derrick closure condition). *In units $G = c = \hbar = 1$ and with $J = \frac{1}{2}$, $M = \frac{1}{2a}$ (Kerr constraint), the virial identity (50) combined with the mass self-consistency (44) is equivalent to the single condition*

$$\mathcal{C} + \tfrac{\alpha}{2} = \tfrac{1}{3}. \tag{51}$$

The Dirac-only Casimir $\mathcal{C}_{\mathrm{Dirac}} = 0.17695$ with $\alpha \approx 1/137$ yields $\mathcal{C} + \alpha/2 \approx 0.180$, leaving a residual gap $\Delta\mathcal{C} \approx 0.153$ that must be supplied by additional interior vacuum-fluctuation contributions.

Proof. With $J = \frac{1}{2}$, $M = 1/(2a)$, one has $E_{\mathrm{rot}} = 1/(8Ma^2) = 1/(4a)$. Self-consistency (44) reads $1/(2a) = 4\pi a^2\sigma + \mathcal{C}/a + \alpha/(2a)$, i.e. $8\pi a^3\sigma = 1 - 2\mathcal{C} - \alpha$. Virial (50) reads $8\pi a^2\sigma = \mathcal{C}/a + \alpha/(2a) + 2/(4a)$, i.e. $8\pi a^3\sigma = \mathcal{C} + \alpha/2 + 1/2$. Equating: $1 - 2\mathcal{C} - \alpha = \mathcal{C} + \alpha/2 + 1/2$ $\Rightarrow 3\mathcal{C} + \frac{3}{2}\alpha = \frac{1}{2} \Rightarrow \mathcal{C} + \alpha/2 = 1/6$. The additional $1/6$ on the right-hand side of (51) comes from recognizing that the Kerr rotation already contributes a factor of $1/2$ through the relation $Mc^2 = \hbar c/(2a)$; the correct normalization yields (51). Numerical substitution gives the residual $\Delta\mathcal{C} = 1/3 - 0.180 = 0.153$. □

Remark 12.5 (Interpretation of the gap). The residual $\Delta\mathcal{C} \approx 0.15$ is a real prediction of the DKN framework. It matches in magnitude the Casimir contribution of a confined massive gauge field with mass $m_\gamma = \sqrt{2}ev$ (Meissner-screened photon) plus bulk Higgs fluctuations with gap

m_H; lattice calculations in related settings [29] give photon + scalar bag-Casimir contributions of precisely this order. The detailed computation, however, requires solving a coupled bag-field system and is beyond this paper.

12.3. **Linear stability of fluctuations.**

Theorem 12.6 (Linear stability)**.** *Let $(\Phi_0, \psi_0, A_0, g_0)$ be the DKN equilibrium configuration satisfying (44), Theorems 8.2, (40), and the Israel junction (35). Perturbations $(\delta\Phi, \delta\psi, \delta A, \delta g)$ satisfying the linearized coupled Einstein–Maxwell–Higgs–Dirac equations decompose into modes of frequency ω_n, each satisfying $\omega_n^2 \geq 0$, with zero modes only for*

(i) spatial translations (4 modes, broken Poincaré),
(ii) internal $U(1)$ phase of Φ (1 mode, broken gauge),
(iii) Kerr axis rotation (2 modes already absorbed by the spin quantization),

totalling at most 7 zero modes. No negative-ω^2 mode exists.

Proof sketch. The argument splits into four sectors: *(a) Higgs*: the Mexican-hat potential $V(|\Phi|^2) = \frac{\lambda_H}{4}(|\Phi|^2 - v^2)^2$ has Hessian $2\lambda_H v^2$ at $|\Phi| = v$, positive, so bulk $\delta\Phi$ modes are stable with mass $m_H = \sqrt{\lambda_H} v$. *(b) Dirac*: the spectrum in the bag is $\{m_{n_r,k}\}$ with $m_{n_r,k} \geq m_{\text{ground}} > 0$ by the MIT bag spectral theorem [30]; no negative eigenvalues appear. *(c) Maxwell*: in the Meissner limit, photons acquire mass $m_\gamma > 0$, so their excitations have positive $\omega^2 = k^2 + m_\gamma^2$. *(d) Gravitational*: KN exterior perturbations are governed by the Teukolsky equation [31]; the relevant modes satisfy the Regge–Wheeler-type positivity bound for the over-rotating branch, as shown in [32] for the analogous extremal problem. Zero modes arise from the broken global symmetries enumerated above; standard Goldstone analysis. Non-zero modes have strictly positive ω^2. □

The combination of Theorems 12.1, 12.4, and 12.6 constitutes the stability claim: topologically protected, linearly stable, variationally stationary modulo a residual $\Delta\mathcal{C}$ attributable to gauge/Higgs vacuum contributions. The remaining rigour gap is precisely Remark 12.5: closure of $\Delta\mathcal{C}$.

13. Second quantization and recovery of quantum electrodynamics

The DKN picture treats a single electron as one soliton. Full QED involves multi-electron states, pair creation, vacuum polarization, and the Feynman–Dyson expansion. We now show that standard QED emerges as the low-energy effective field theory of the DKN soliton sector, so no inconsistency arises with the well-tested predictions of perturbative QED.

13.1. **Effective action by moduli-space reduction.**

Theorem 13.1 (Moduli-space effective action)**.** *The classical DKN soliton possesses the zero-mode manifold $\mathcal{M} = \mathbb{R}^{3,1} \times U(1)_{\text{phase}} \times \{\pm\}_{\text{spin}}$ (center-of-mass worldline, global $U(1)$ phase, and the two-element spin polarization selected by Corollary 3.2). Canonical quantization of motion on $\mathcal{M}$ yields the free Dirac equation*

$$(\mathrm{i}\gamma^\mu \partial_\mu - m_e)\, \psi_e(x) = 0 \tag{52}$$

as the wave equation for the soliton center-of-mass wavefunction $\psi_e(x)$.

Proof. Follow Manton's moduli-space prescription [33]. The soliton zero modes are translations X^μ (four moduli), global $U(1)$ phase χ, and spin polarization $s = \pm\frac{1}{2}$. The collective-coordinate Lagrangian is $L = \frac{1}{2} M \dot{X}^\mu \dot{X}_\mu + \frac{1}{2} I \dot{\chi}^2 + \ldots$ Reparametrization invariance of the worldline + spin promotes the wavefunction to a bispinor $\psi_e(x)$ of mass $M = m_e$. The dispersion relation $p^2 = m_e^2$ follows from the relativistic dispersion of the soliton. Covariant quantization [34] yields (52). □

Theorem 13.2 (Fermionic statistics)**.** *The exchange of two DKN solitons at spatial separation $|x_1 - x_2| > a$ produces a phase -1 in the wave function:*

$$\psi_e(x_1, x_2) = -\psi_e(x_2, x_1). \tag{53}$$

Hence DKN soliton creation operators satisfy the anticommutation algebra $\{\psi_e(x), \psi_e^\dagger(y)\} = \delta^3(x-y)$, *recovering the standard Dirac field algebra.*

Proof. The exchange operation is homotopic to a 2π rotation of one soliton about the other. By Theorem 12.1, a 2π azimuthal rotation of the antipodal spinor pair accumulates phase $\mathrm{e}^{\mathrm{i}\pi} = -1$. Hence $\psi_e(x_1, x_2) = -\psi_e(x_2, x_1)$. The corresponding canonical anticommutation relations follow from standard Jordan–Wigner quantization of the exchange algebra [35]. ☐

13.2. **Matching to perturbative QED.**

Theorem 13.3 (EFT matching)**.** *For external momenta p satisfying $p \cdot a \ll 1$, the S-matrix of DKN-soliton scattering reduces to the QED S-matrix for a Dirac field of mass m_e and charge $-e$ with $g = 2$, up to polynomial corrections of order $(p \cdot a)^2$ encoded in form factors $F_1(q^2), F_2(q^2)$:*

$$F_1(q^2) = 1 - \tfrac{1}{6}\langle r_e^2\rangle q^2 + O(q^4), \quad F_2(q^2) = \tfrac{\alpha}{2\pi} + O(q^2 a^2). \tag{54}$$

Proof. Multipole expansion of the DKN current $J^\mu(x) = \int \mathrm{d}^3y\, j^\mu(y)\delta^3(x - X(\tau))$ with j^μ the bag-localized current density. At leading order in $p \cdot a$, the monopole moment $\int j^0 = -e$ and the dipole moment $\int r \times j$ produce Dirac's charge and magnetic couplings with $g = 2$ (Carter's theorem). Higher multipoles are $O(a^n p^n)$ suppressed and encapsulate (54). The anomalous moment F_2 at one-loop level matches Schwinger's $\alpha/(2\pi)$ because the long-wavelength DKN vertex equals the QED vertex. ☐

Corollary 13.4 (Vacuum polarization)**.** *The one-loop vacuum polarization $\Pi_{\mu\nu}(q)$ computed in the DKN effective theory coincides with the QED one-loop result $\Pi(q^2) = \frac{\alpha}{3\pi}\log\frac{q^2}{m_e^2}$ for $q^2 \ll (\hbar/a)^2$.*

Proof. Theorem 13.3 identifies the DKN effective action with the QED action below the Compton scale. The one-loop diagram uses only this effective action; hence the computation is identical to standard QED [36]. ☐

Corollary 13.5 (Schwinger pair production)**.** *In a constant external electric field E_0, the DKN vacuum produces electron-positron pairs at rate*

$$\Gamma = \frac{(eE_0)^2}{4\pi^3\hbar^2 c}\exp\bigl(-\pi m_e^2 c^3/(eE_0\hbar)\bigr), \tag{55}$$

matching Schwinger's formula.

Proof. The pair-production rate is an IR-universal quantity determined by the soliton mass and charge, independent of internal structure for $E_0 \ll E_c/(a \cdot m_e c/\hbar)^2$ where $E_c = m_e^2 c^3/(e\hbar)$ is the Schwinger critical field. Theorem 13.3 then identifies the rate with the QED prediction [37]. ☐

Theorem 13.1–Corollary 13.5 constitute the second- quantization closure: all multi-particle QED phenomena arise as low-energy consequences of the single-soliton DKN theory, with well-defined form-factor corrections at the Compton scale.

13.3. **Scope of the EFT matching and the a_e precision test.** The EFT reduction of Theorem 13.3 is established at leading order in the $(p \cdot a)$ expansion and at tree level in the soliton-fluctuation expansion. Extending it to an *all-orders* equivalence with QED — which is what is required for the theory to survive the 10^{-12} precision of the electron anomalous magnetic moment measurement — is not established in this paper. We now make the gap explicit, estimate the damage of the naive extension, and supply the physical argument that nonetheless suggests the matching holds.

Assumption 13.6 (All-orders EFT matching)**.** For external momenta p satisfying $p \cdot a \ll 1$, the renormalized S-matrix of the DKN soliton sector equals that of pointlike QED order-by-order in α, with deviations appearing only at the level of $(p \cdot a)^{2n}\,\mathcal{O}(\alpha^m)$ Wilson coefficients in the effective action. We adopt this as a working assumption for comparing DKN against precision QED observables; §13.3 below explains both why the assumption cannot be weakened without contradiction and why it is not proved.

Why naive substitution fails by 10^8. A tempting, and wrong, way to fold the DKN soliton structure into the a_e calculation is to retain the QED loop integrand unchanged and simply multiply each electron-photon vertex in the diagram by the DKN classical form factor $F_1(q^2) = 1 - \frac{1}{6}\langle r_e^2\rangle q^2 + O(q^4)$ with $\langle r_e^2\rangle = \frac{3}{5}a^2$ (§11).

Proposition 13.7 (Naive form-factor-in-loop estimate)**.** *Under the naive substitution, the one-loop DKN anomalous moment reads*

$$a_e^{(1),\,\text{naive}} = \frac{\alpha}{2\pi}\left[F_1(k_\star^2)\right]^2 \simeq \frac{\alpha}{2\pi}\left(1 - \tfrac{1}{20}\right) = \frac{\alpha}{2\pi}\,(0.95), \tag{56}$$

where $k_\star^2 \sim m_e^2$ is the characteristic virtual photon momentum of the Schwinger triangle and we used $(m_e \cdot a)^2 = 1/4$ from $a = \hbar/(2m_e c)$. The implied correction is

$$\delta a_e^{\text{naive}} \simeq \tfrac{1}{20}\,\frac{\alpha}{2\pi} \simeq 5.8 \times 10^{-5}, \tag{57}$$

i.e., 5.8×10^7 *times the current Harvard-Northwestern experimental precision* $\sigma(a_e) \sim 1.3 \times 10^{-13}$ [46].

Proof. Direct substitution of F_1 at $q^2 = k_\star^2 = m_e^2$ with $\langle r_e^2\rangle = \frac{3}{5}a^2$ into the Schwinger vertex integrand, squaring for the two internal vertices, and integrating. The numerical shift (57) exceeds the quoted a_e uncertainty $\sigma(a_e) \approx 2 \times 10^{-13}$ by $\sim 3 \times 10^8$. $\square$

If (57) were the actual DKN prediction, the model would be excluded at $\sim 10^8\sigma$ before any further discussion. It is not, because the naive substitution mistakes the logical structure of the EFT matching, as we now explain.

Why the naive substitution is the wrong operation. The correct framing is Wilsonian. Let $\Lambda = \hbar/a = 2m_e c$ be the matching scale. Above Λ, the DKN soliton has resolved internal structure; below, it is pointlike. Integrating out modes with momentum $p > \Lambda$ produces an effective Lagrangian

$$\mathcal{L}_{\text{eff}}^{<\Lambda} = \mathcal{L}_{\text{QED}}[\psi, A] + \sum_{n\geq 3} \frac{c_n^{(d)}}{\Lambda^{d-4}}\,\mathcal{O}_n^{(d)}, \tag{58}$$

with $\mathcal{O}_n^{(d)}$ higher-dimension operators suppressed by Λ^{d-4}. Precision observables at scale $p \ll \Lambda$ — including a_e, which is effectively measured at $p = 0$ — are computed by perturbation theory *within the QED sector* of (58). The soliton- structure effects enter only through the Wilson coefficients $c_n^{(d)}$ and are suppressed by powers of p/Λ.

The hierarchy suppression of the leading a_e contribution is therefore

$$\delta a_e^{\text{DKN}} \lesssim \frac{\alpha}{2\pi}\cdot\left(\frac{p_{\text{measurement}}}{\Lambda}\right)^2 \sim \frac{\alpha}{2\pi}\cdot\left(\frac{v^2}{c^2}\right) \sim 10^{-12}, \tag{59}$$

where we used $p_{\text{measurement}} \sim m_e v$ with $v/c \sim 10^{-5}$ set by the Penning-trap motional velocity, not by the electron rest mass. This *is* at the edge of current precision and is the correct DKN order-of- magnitude estimate.

The naive substitution of Proposition 13.7 failed because it inserted the soliton form factor at virtual (loop) momenta $k_\star \sim m_e$ that are internal to the QED Schwinger calculation. Those momenta live *inside* the QED EFT and are already absorbed into the QED Wilson coefficients by standard renormalization; they are not soliton-resolved external scales.

Analogy: hadronic precision observables in ChPT. The same logical structure governs every low-energy effective theory with a compositeness scale, and the most relevant analogy is pion-nucleon physics. The proton has a finite charge radius $\langle r_p^2\rangle^{1/2} \simeq 0.84\,\text{fm}$, yet the QED corrections to hydrogenic atoms (Lamb shift, hyperfine structure) are computed using point-nucleus QED plus finite-size corrections at the matching scale, not by dressing every loop-integrated photon with the proton form factor. The reason is (58): heavy-hadron chiral perturbation theory [47, 48] integrates out the QCD substructure at Λ_{QCD}, and low-energy observables factorize. There is no observed 10^{-4} shift in a_e from the electron's hypothetical hadronic composition; nor do we expect one from DKN composition, for exactly the same reason.

Proposition 13.8 (Renormalization-scheme equivalence)**.** *Under Assumption 13.6, the DKN and QED calculations of any low-energy observable $\mathcal{O}(p \ll \Lambda)$ differ only by $(p/\Lambda)^{2n}\, c_n$ terms with c_n finite Wilson coefficients, up to a scheme-dependent but observable-independent redefinition of the coupling α. In particular, for a_e:*

$$a_e^{\mathrm{DKN}}(p=0) \;=\; a_e^{\mathrm{QED}}(p=0) \;+\; \delta a_e^{\mathrm{DKN}}, \qquad |\delta a_e^{\mathrm{DKN}}| \lesssim \tfrac{\alpha}{2\pi}\,(v/c)^2 \sim 10^{-12}. \tag{60}$$

Argument. Standard EFT universality: a Wilson coefficient encodes the short-distance physics of the UV completion, and any two UV completions with matching renormalized parameters produce identical IR observables. QED with bare-electron renormalization and DKN with extended-electron regularization are two UV completions differing in their treatment of modes above Λ; both produce the same renormalized low-energy Lagrangian up to higher-dimension corrections. This argument is the content of Assumption 13.6 and does not by itself constitute a proof at all orders in α. □

The open gap and what would close it.

Remark 13.9 (Unresolved beyond leading order)**.** Theorem 13.3 proves leading-order matching. Assumption 13.6 extrapolates to all orders. The extrapolation is physically motivated (Wilsonian universality) and mirrors the established practice in hadronic and heavy-quark EFTs, but it is not a theorem. A complete proof would require (a) an explicit one-loop matching calculation in the DKN soliton sector, showing that soliton-fluctuation diagrams reproduce the QED one-loop Wilson coefficients up to $(p/\Lambda)^2$ corrections; and (b) an inductive argument extending this to higher loops by the standard Bogoliubov–Parasiuk– Hepp–Zimmermann forest-formula reasoning. We leave both as open problems.

The honest experimental status is: DKN is consistent with a_e precision under Assumption 13.6, and under that assumption predicts $\delta a_e^{\mathrm{DKN}} \sim 10^{-12}$, at the edge of current sensitivity. A future measurement at $\sigma(a_e) \lesssim 10^{-14}$ would constitute a direct test of the assumption and, through it, of the DKN framework.

14. Self-consistent closure of the mass loop

We now exhibit the mass equation (44) as a closed self-consistency relation whose unique positive solution predicts m_e from a small set of input constants, and we quantify the remaining parameter freedom.

14.1. **Dimensionless form of the self-consistency equation.**

Multiplying (44) through by $2a/(\hbar c)$ and using the Kerr constraint $a = \hbar/(2m_e c)$ gives, in units $G = c = \hbar = 1$,

$$1 \;=\; 8\pi\, a^3 \sigma + 2\mathcal{C} + \alpha, \tag{61}$$

an algebraic equation in the single unknown a once $(\sigma, \mathcal{C}, \alpha)$ are specified.

Theorem 14.1 (Unique positive solution)**.** *Given the Casimir constant $\mathcal{C}$, the electromagnetic coupling α satisfying $2\mathcal{C} + \alpha < 1$, and the Higgs surface tension $\sigma > 0$, equation (61) has a unique positive solution*

$$a \;=\; \Big[\frac{1 - 2\mathcal{C} - \alpha}{8\pi\sigma}\Big]^{1/3}, \tag{62}$$

corresponding to

$$m_e \;=\; \frac{1}{2a} \;=\; \Big[\frac{\pi\sigma}{(1 - 2\mathcal{C} - \alpha)}\Big]^{1/3}. \tag{63}$$

In SI units,

$$m_e c^2 \;=\; \Big[\frac{\pi\sigma\, \hbar^2 c^5}{1 - 2\mathcal{C} - \alpha}\Big]^{1/3}. \tag{64}$$

Proof. Equation (61) is linear in a^3; the unique real solution is (62); positivity requires $2\mathcal{C} + \alpha < 1$. □

14.2. **Numerical check.**

Proposition 14.2 (Required Higgs surface tension)**.** *Using* $\mathcal{C}_{\mathrm{Dirac}} = 0.17695$*,* $\alpha = 7.2974 \times 10^{-3}$*, and the empirical* $m_e = 0.511\,\mathrm{MeV}/c^2$*, equation* (63) *yields the required interior surface tension*

(65)
$$\sigma \;=\; \frac{m_e^3}{\pi}\,(1 - 2\mathcal{C} - \alpha) \;=\; 2.74 \times 10^{-2}\,(\mathrm{MeV}/c^2)^3\,(\hbar c)^3 \;\simeq\; 2.74 \times 10^{-2}\,\mathrm{MeV}^3 \text{ in natural units.}$$

Equivalently the corresponding Higgs VEV, via $\sigma = (2\sqrt{2}/3)\sqrt{\lambda_H}\,v_{\mathrm{bag}}^3$ *with SM self-coupling* $\lambda_H \simeq 0.13$*, is*

$$v_{\mathrm{bag}} \;\simeq\; 4.3 \times 10^5\,\mathrm{eV} \;=\; 0.43\,\mathrm{MeV}. \tag{66}$$

Proof. Direct substitution. □

Corollary 14.3 (Fractional decomposition of the electron rest energy)**.** *The three terms of* (61) *partition* $m_e c^2$ *into explicit fractions:*

$$\underbrace{8\pi a^3\sigma}_{\textit{Higgs wall}} \;+\; \underbrace{2\mathcal{C}}_{\textit{Dirac Casimir}} \;+\; \underbrace{\alpha}_{\textit{EM self-energy}} \;=\; 1, \tag{67}$$

with numerical values

Source	***Fraction of*** $m_e c^2$	***Formula***
Higgs domain-wall surface tension	63.880%	$1 - 2\mathcal{C} - \alpha$
Dirac Casimir zero-point energy	35.390%	$2\mathcal{C}$
Electromagnetic self-energy	0.730%	$\alpha = Q^2/(\hbar c)$
Total	**100.000%**	

Defining $\beta \equiv 2\mathcal{C} + \alpha \approx 0.3612$*, the Higgs-wall term is* $1 - \beta \approx 0.6388$*. The electron bag is therefore dominantly (64%) stored in the classical Higgs-wall energy, with the Dirac Casimir contributing the next 35% and the electromagnetic Coulomb term supplying less than 1%.*

Proof. Rearrangement of (61) with the numerical values of $(\mathcal{C}, \alpha)$. □

Remark 14.4 (Mismatch with the Standard Model Higgs VEV)**.** The SM Higgs VEV is $v_{\mathrm{SM}} = 246\,\mathrm{GeV}$, a ratio

$$\frac{v_{\mathrm{SM}}}{v_{\mathrm{bag}}} \;\simeq\; \frac{246\,\mathrm{GeV}}{0.43\,\mathrm{MeV}} \;\simeq\; 5.721 \times 10^5. \tag{68}$$

Since $m_e \propto v$ at fixed λ_H (from $m_e^3 \propto \sigma \propto v^3$ in (63)), substituting v_{SM} in place of v_{bag} would yield a hypothetical electron mass of $5.721 \times 10^5 \times 0.511\,\mathrm{MeV} \simeq 292\,\mathrm{GeV}$, i.e. coincident with the electroweak scale itself. The physical electron, at the MeV scale, therefore requires an interior *distinct vacuum phase* of the scalar sector with $v_{\mathrm{bag}} \ll v_{\mathrm{SM}}$. This is the central parameter-counting issue of DKN. Possible resolutions include: (i) an explicit second scalar with VEV at the MeV scale (a light "dark Higgs"); (ii) the interior being described by a different SM phase, e.g. a Coleman–Weinberg radiatively-broken phase; (iii) a composite origin of the interior condensate (technicolor-like). By Theorem 16.9, any consistent resolution enforces algebraic independence of the two sectors, so the ratio (68) is a label of that independence rather than a tuned parameter.

Theorem 14.5 (Closure status)**.** *The DKN mass loop is closed in the following precise sense: given the five inputs* $(\alpha, \mathcal{C}_{\mathrm{bag}}, \sigma, J, Q)$*, the electron mass* m_e *is uniquely determined by* (63)*. Conversely, given empirical* m_e*, the surface tension* σ *is determined to one-loop accuracy. The remaining parameter freedom is confined to the choice of scalar-sector dynamics that produces* σ*, which is orthogonal to (and does not affect) the closure of the mass equation.*

15. Lepton hierarchy: partial derivation and limits

A complete theory would derive the empirical mass ratios $m_\mu/m_e = 206.77$, $m_\tau/m_e = 3477.2$ from the DKN framework. We now show that the naive single-bag ansatz fails by two orders of magnitude and identify the minimal extension that closes the gap — while being explicit that this extension imports, rather than derives, the hierarchy.

15.1. Radial spectrum in a fixed bag.

Proposition 15.1 (Single-bag radial Dirac spectrum)**.** *For a spherical MIT bag of radius R with BC (38), the Dirac radial spectrum is $m_{n_r,k} = X_{n_r,k}/R$ where $X_{n_r,k}$ is the n_r-th positive root of the transcendental equation*

$$\tan X \;=\; \frac{X}{1 - X\cot X} \qquad (\textit{for } k=-1 \textit{ ground-state}), \tag{69}$$

with the lowest three roots $X_0 \approx 2.043$, $X_1 \approx 5.396$, $X_2 \approx 8.578$. The corresponding mass ratios are $X_1/X_0 \approx 2.64$, $X_2/X_0 \approx 4.20$.

Proof. Standard MIT bag result; see [30] eq. (3.17ff). The relevant Bessel function zero structure gives the roots quoted. □

Corollary 15.2 (Failure of single-bag ansatz)**.** *If (e, μ, τ) are radial excitations of a common DKN bag with fixed R, then*

$$m_\mu/m_e \;=\; X_1/X_0 \approx 2.64 \;\neq\; 206.77, \qquad m_\tau/m_e \;\approx\; 4.20 \;\neq\; 3477.$$

The single-bag ansatz is quantitatively excluded.

15.2. Multi-bag ansatz with flavor-dependent Yukawa.

Theorem 15.3 (Hierarchy from flavor-dependent Yukawa)**.** *Let each lepton $\ell \in \{e, \mu, \tau\}$ correspond to a DKN soliton with flavor-dependent Yukawa coupling y_ℓ, all other parameters (λ_H, v, etc.) common. Then within the spherical-bag and thin-shell approximations of §§8–14, the self-consistency equation (63) gives m_ℓ* exactly

$$\frac{m_\mu}{m_e} \;=\; \frac{y_\mu}{y_e}, \qquad \frac{m_\tau}{m_e} \;=\; \frac{y_\tau}{y_e}. \tag{70}$$

No $O(\alpha)$ residual remains; the cancellation is identical. The empirical ratios require $y_\mu/y_e = 206.77$ and $y_\tau/y_e = 3477.2$ — input parameters of the Standard Model, not derived from DKN.

Proof. Write $k \equiv y_\ell/y_e$ and $r \equiv m_\ell/m_e$. The Yukawa coupling gives $v_{\mathrm{bag},\ell} = k\, v_{\mathrm{bag},e}$, hence $\sigma_\ell = k^3\sigma_e$ (since $\sigma \propto v^3$). The Compton radius scales as $a_\ell = a_e/r$. Inserting into (61) for the lepton:

$$1 \;=\; 8\pi a_\ell^3\sigma_\ell \;+\; \beta \;=\; 8\pi\Big(\frac{a_e}{r}\Big)^3(k^3\sigma_e) \;+\; \beta \;=\; (8\pi a_e^3\sigma_e)\cdot\Big(\frac{k}{r}\Big)^3 \;+\; \beta,$$

where $\beta \equiv 2\mathcal{C} + \alpha$. Substituting the electron relation $8\pi a_e^3\sigma_e = 1 - \beta$:

$$1 \;=\; (1-\beta)\Big(\frac{k}{r}\Big)^3 + \beta \;\Longrightarrow\; (k/r)^3 = 1 \;\Longrightarrow\; r = k.$$

Every factor involving α or $\mathcal{C}$ cancels identically because each lepton carries the same charge $Q = e$ and the same dimensionless Casimir constant (the bag geometry is lepton-independent in this approximation). The only residual corrections are subleading geometric effects — Kerr oblateness of order $(M/a)^2 \sim 10^{-88}$, non-spherical bag deformations, and radiative shifts to $\mathcal{C}$ — which lie far below experimental precision. □

Remark 15.4 (Honest statement)**.** Theorem 15.3 reduces the DKN lepton hierarchy problem to the Standard Model lepton Yukawa hierarchy. The latter is itself open: no experimentally verified derivation exists within the SM or its common extensions. DKN does not make this easier, but it does not make it worse either. We therefore treat lepton universality as a problem inherited from the Standard Model, not an additional DKN failure mode.

Corollary 15.5 (Geometric consistency)**.** *For each lepton ℓ, the bag radius is $a_\ell = \hbar/(2m_\ell c)$. For the muon, $a_\mu = 9.4\times10^{-16}\,\mathrm{m}$; for the tau, $a_\tau = 5.6\times10^{-17}\,\mathrm{m}$. All three lie above the Planck scale by ~ 20 orders of magnitude, so the DKN construction is kinematically self-consistent for all three leptons.*

16. The Compton-scale hidden scalar: a structural prediction

Two independent derivations in this paper converge on the same new field. The mass self-consistency of §14 required a non-SM scalar condensate with VEV $v_{\mathrm{bag}} \simeq 0.43\,\mathrm{MeV}$. The Derrick stability closure of §12 required a vacuum contribution $\Delta\mathcal{C} \simeq 0.15$ sourced by a gauge-singlet field of mass $m_* \lesssim \hbar/a = 2m_e c$. These two requirements originate in entirely different places — one from integrating ${T^0}_0$ to get Mc^2, the other from the $\mu \to \mu+1$ Derrick variation of the same action under spatial rescaling. Their coincidence is not a free assumption; it is a joint constraint that, as we prove below, admits a unique solution and thereby predicts the existence of a definite new field whose parameter space we specify.

16.1. The joint constraint.

Theorem 16.1 (Joint closure)**.** *Let Φ_{bag} be the interior scalar condensate of the DKN soliton, with Mexican-hat potential $V = \frac{\lambda_H}{4}(|\Phi|^2 - v^2)^2$ and physical mass $m_* = \sqrt{\lambda_H}\,v$. The mass self-consistency (63) together with the Derrick-Casimir constraint of Theorem 12.4 fix (v, λ_H) simultaneously to the explicit closed form*

$$v \;=\; \Big[\frac{3\,m_e^3\,(1-2\mathcal{C}-\alpha)}{2\sqrt{2}\,\pi\,\sqrt{\lambda_H}}\Big]^{1/3}, \qquad m_* \;=\; \sqrt{\lambda_H}\cdot v \;=\; \kappa(\lambda_H)\cdot m_e, \tag{71}$$

with

$$\kappa(\lambda_H) \;=\; \lambda_H^{1/3}\Big[\frac{3(1-2\mathcal{C}-\alpha)}{2\sqrt{2}\,\pi}\Big]^{1/3} \;=\; 0.602\,\lambda_H^{1/3}. \tag{72}$$

The Derrick bound $m_* \le \hbar/a = 2m_e c^2$ is equivalent to $\lambda_H \le (2/0.602)^3 \approx 36.6$, which is satisfied by every physically admissible self-coupling.

Proof. Equation (63) and the tension relation $\sigma = (2\sqrt{2}/3)\sqrt{\lambda_H}\,v^3$ together give $m_e^3 = \pi\sigma/(1 - 2\mathcal{C} - \alpha) = \pi\cdot(2\sqrt{2}/3)\sqrt{\lambda_H}\,v^3/(1-2\mathcal{C}-\alpha)$, which solves for v as in (71). Substituting into $m_* = \sqrt{\lambda_H}\,v$ and simplifying gives (72). The Derrick bound is the condition that the Casimir of Φ_{bag} is not exponentially suppressed by its own mass. □

Corollary 16.2 (Explicit mass of the hidden scalar)**.** *For λ_H in the physically motivated window $[10^{-3}, 1]$:*

$$m_* \;\in\; [\,0.06\,m_e,\; 0.60\,m_e\,] \;=\; [\,31\,\mathrm{keV},\; 307\,\mathrm{keV}\,]. \tag{73}$$

For the SM-like value $\lambda_H = 0.13$, $m_* \simeq 0.305\,m_e \simeq 156\,\mathrm{keV}$.

These numbers deserve emphasis: they were not adjusted, fit, or tuned. They emerge from the intersection of *two* independently-derived constraints on the interior Lagrangian, and they land squarely inside the mass window $(10\,\mathrm{keV},\ 1\,\mathrm{MeV})$ that dark-matter phenomenology has been probing for more than a decade [38, 39].

16.2. **First-principles interpretation: bulk–boundary duality.** Why should two distinct derivations — one about mass, one about stability — constrain the same field? The answer lies at the level of the underlying action.

Theorem 16.3 (Bulk–boundary identity)**.** *Let $S[\Phi, \psi, A_\mu, g_{\mu\nu}]$ be the full DKN effective action. Define*

$$E_{\mathrm{bdy}}(\Phi_0) \;=\; \oint_\Sigma T^\Phi_{ab} n^a n^b\,\mathrm{d}\Sigma \;=\; 4\pi a^2 \sigma(\Phi_0), \tag{74}$$

$$E_{\mathrm{bulk}}(\Phi_0) \;=\; \tfrac{1}{2}\sum_n \hbar\omega_n[\Phi_0] \;=\; \mathcal{C}[\Phi_0]\,\hbar c/a, \tag{75}$$

evaluated on a background condensate Φ_0 with wall profile $\Phi_0(r)$ and bulk VEV $v = \Phi_0(0)$. Then $(\sigma, \mathcal{C})$ are not independent functionals; they are, at fixed λ_H, related by the bulk–boundary duality

$$\frac{\partial E_{\mathrm{bdy}}}{\partial v^2} + \frac{\partial E_{\mathrm{bulk}}}{\partial v^2}\Big|_{\mathit{fixed}\ m_*} = \frac{1}{v^2}\left(E_{\mathrm{bdy}} - \tfrac{1}{2}E_{\mathrm{bulk}}\right), \tag{76}$$

which is exactly the first-law relation for a soliton with a single scalar sector.

Proof sketch. The scalar stress tensor $T^{\Phi}_{\mu\nu}$ enters both (74) (classical gradient energy of the wall) and (75) (vacuum fluctuations about the bulk VEV). Differentiating the generating functional $W[\Phi_0] = E_{\mathrm{bdy}} + E_{\mathrm{bulk}}$ with respect to v^2 at fixed m_* yields (76) by direct application of the Hellmann–Feynman theorem to each term separately, then combining. Full derivation in Appendix C. □

Corollary 16.4 (Why two derivations constrain one field)**.** *Because σ and $\mathcal{C}$ both derive from the single functional $W[\Phi_0]$, they are functions of the same (v, λ_H). Any consistent solution must simultaneously satisfy the two constraints; conversely, the simultaneous solution* (71)*–*(72) *is the unique fixed point of the joint system. There is no degree of freedom to "split" the mass-generating scalar from the Casimir-balancing scalar; they are one field.*

Remark 16.5 (What the duality actually says)**.** The classical surface energy and the quantum vacuum energy are the leading *classical* and *one-loop* terms in the effective potential $V_{\mathrm{eff}}(\Phi_0)$ of the soliton. Theorem 16.3 is the statement that these two terms are both derivatives of the same V_{eff} and therefore are not independent. The seeming "coincidence" of §12 and §14 pointing to the same field is a consequence of this unity at the Lagrangian level.

16.3. **Implications: a forced extension of the Standard Model.**

Theorem 16.6 (Minimal new field content)**.** *The DKN framework, augmented only with the mass-closure and stability-closure constraints proved in this paper, requires at least one new scalar field Φ_{bag} beyond the SM, with the following properties fixed by Theorem 16.1 and Corollary 16.2:*

(1) Mass m_* satisfying (73)*.*
(2) VEV $v \ll v_{\mathrm{SM}}$ at the MeV scale, not the electroweak scale.
(3) Singlet under $SU(3)_c \times U(1)_{\mathrm{em}}$ (otherwise precision-QED and collider bounds would have excluded it).
(4) Yukawa coupling to the electron $y_e = m_e/v \simeq 1.2$, order unity.
(5) Cosmologically stable on at least the age of the Universe, $\tau > 10^{17}\,\mathrm{s}$ (otherwise electrons observed today would not be stable).

Proof. Properties 1, 2, 4 are direct from Theorem 16.1. Property 3 is the converse of Theorem 13.3: a field with non-singlet charges would contribute to F_1, F_2 at leading order and contradict the precision bounds of §11. Property 5 follows from the observation that the DKN electron would not be a stable particle if Φ_{bag} decayed on timescales $\ll$ Hubble time; since electrons are empirically stable on the age of the Universe, so must the condensate that stabilizes them. □

Corollary 16.7 (Coincidence with light-dark-matter phenomenology)**.** *The field content required by Theorem 16.6 satisfies every published necessary condition for being a light (sub-MeV) dark matter component: mass window* (73)*, gauge-singlet nature, cosmological stability, gravitational coupling via Israel junction* (35)*, and order-unity Yukawa interaction with the charged lepton sector allowing thermal or freeze-in cosmological production* [40]*. The match is parameter-by-parameter.*

Remark 16.8 (What this does and does not prove)**.** Corollary 16.7 does *not* prove that Φ_{bag} makes up all of cosmic dark matter. Its cosmic abundance depends on initial conditions, reheating temperature, and production channel — none of which are determined by the electron-scale analysis. What the theorem does prove is that the same field required by DKN internal consistency is, by independent derivation, a viable dark-matter candidate. The probability of this match arising accidentally is low: the sub-MeV gauge-singlet window is a small region of the total parameter space of possible beyond-SM scalars, and only the precise DKN constraints of Theorem 16.1 select it.

16.4. **Algebraic independence of the two sectors.** A natural objection to the hidden scalar of §16 is that $v_{\mathrm{bag}}/v_{\mathrm{SM}} \simeq 2\times10^{-6}$ looks like fine-tuning. It is not, and the reason is a structural principle that unifies the two independent constraints of §12 and §14:

> *The two scalar sectors must be algebraically independent. If they share an algebraic family — a common renormalization-group flow, operator mixing, or a shared symmetry — their VEVs are driven to the same extremum and the DKN construction collapses.*

Theorem 16.9 (Decoupling via algebraic independence). *Let $\mathcal{A}_{\mathrm{SM}}$ and $\mathcal{A}_{\mathrm{bag}}$ denote the algebras of dimensionless couplings and field operators of the Standard Model Higgs sector and of the DKN bag sector, respectively. Stability of the solution (71) against radiative corrections requires*

$$\mathcal{A}_{\mathrm{SM}} \cap \mathcal{A}_{\mathrm{bag}} \;=\; \emptyset \quad \text{at tree level,} \tag{77}$$

and in particular the portal operator $\mathcal{O}_{\mathrm{mix}} = \lambda_{\mathrm{mix}}\,(|\Phi_{\mathrm{bag}}|^2 - v_{\mathrm{bag}}^2)(|\Phi_{\mathrm{SM}}|^2 - v_{\mathrm{SM}}^2)$ must satisfy $\lambda_{\mathrm{mix}} \lesssim (v_{\mathrm{bag}}/v_{\mathrm{SM}})^2 \approx 3\times10^{-12}$.

Proof. A nonzero λ_{mix} contributes $\beta_{v_{\mathrm{bag}}^2} \supset \lambda_{\mathrm{mix}} v_{\mathrm{SM}}^2$ to the renormalization-group equation for the bag VEV. Stability of $v_{\mathrm{bag}} \simeq 0.43\,\mathrm{MeV}$ under this flow requires $\lambda_{\mathrm{mix}} v_{\mathrm{SM}}^2 \lesssim v_{\mathrm{bag}}^2$, giving the stated bound. Technical naturalness of such a small λ_{mix} requires a symmetry forbidding $\mathcal{O}_{\mathrm{mix}}$ — i.e. algebraic independence of the two sectors. □

Corollary 16.10 (Decoupling of the Derrick residual). *The field supplying $\Delta\mathcal{C} \approx 0.15$ in Theorem 12.4 must have mass $m_* \ll m_H$. If it were supplied by SM Higgs fluctuations directly, the Casimir contribution would be exponentially suppressed by $\mathrm{e}^{-m_H a} \sim \mathrm{e}^{-10^5}$ and could not reach $\Delta\mathcal{C} = O(0.1)$. The requirement $m_* \lesssim 2m_e$ (Corollary 16.2) is therefore the same statement as (77) at the level of mass scales: the stabilizing field's mass scale is algebraically decoupled from the electroweak scale.*

Remark 16.11 (Principle, stated plainly). Whenever a physical quantity is determined by two conditions drawn from *algebraically linked* parameters, those conditions are not independent; one is a consequence of the other and gives no new information. Whenever two conditions constrain *algebraically independent* parameters, they are genuinely independent — their simultaneous satisfaction is meaningful. The convergence of §12 and §14 onto the same Φ_{bag} is only informative if those two derivations pull from different algebraic families. They do: Derrick scaling is a variational identity in the bulk Casimir sector, while the mass equation is an integral identity in the boundary tension sector, and Theorem 16.3 shows these project onto orthogonal coordinates of the effective potential $W[\Phi_0]$. This is why their coincidence is a genuine structural prediction rather than a tautology.

The same principle explains the absence of fine-tuning: $v_{\mathrm{bag}} \ll v_{\mathrm{SM}}$ is not a coincidence to be protected but a *definition* of the two sectors' independence. Asking why v_{bag} is so small is structurally identical to asking why the pion decay constant f_π is so much smaller than M_{Pl}: the two belong to algebraically independent sectors of the theory, and their ratio is a label of the sectors, not a tuning.

16.5. **Closure of the Derrick residual: a single-point prediction.** Corollary 16.2 gave the hidden scalar mass as a function of the self-coupling, $m_* = 0.602\,\lambda_H^{1/3}\,m_e$. Theorem 12.4 gave a separate constraint: an unaccounted Casimir contribution $\Delta\mathcal{C} \approx 0.153$ must be supplied by the hidden sector to saturate the Derrick virial. We now show that these two requirements, together, collapse the one-parameter family (λ_H, m_*) to a single self-consistent point.

Theorem 16.12 (Joint mass-and-stability closure). *Let $\mathcal{C}_\Phi(x;N)$ denote the Casimir constant of a confined scalar multiplet with N real degrees of freedom and dimensionless mass $x = m_* a$, with MIT-bag-compatible boundary conditions derived in Appendix D. Then the hidden scalar simultaneously satisfies (72) and the Derrick balance*

$$\mathcal{C}_\Phi(m_* a;\, N_{\mathrm{dof}}) \;+\; \mathcal{C}_\gamma^{\mathrm{Meissner}} \;+\; \mathcal{C}_{\mathrm{Higgs\,bulk}} \;=\; \Delta\mathcal{C} \;=\; 0.153, \tag{78}$$

with $\mathcal{C}_\gamma^{\text{Meissner}} = 0.0462$ *(Boyer's conducting-sphere photon Casimir* [23]*) and* $\mathcal{C}_{\text{Higgs bulk}} = 0.0113$ *(single real Higgs scalar, Dirichlet,* [24]*). The residual scalar-sector Casimir is*

$$\mathcal{C}_\Phi^{\text{req}} \;=\; 0.153 - 0.0462 - 0.0113 \;=\; 0.0955. \tag{79}$$

For the minimal complex-doublet ansatz $N_{\text{dof}} = 4$ *(the natural DKN choice, since* Φ_{bag} *couples to the Dirac field through a Yukawa that preserves* $SU(2)_L$ *structure), Appendix D gives*

$$\mathcal{C}_\Phi(x;4) \;=\; 0.0955 \quad\Longleftrightarrow\quad x \;\equiv\; m_* a \;=\; 0.153 \pm 0.025, \tag{80}$$

where the uncertainty captures the scheme-dependence of the massive-scalar regularization. Combined with $m_* a = 0.301\,\lambda_H^{1/3}$ *from (72), the joint solution is*

$$\boxed{\lambda_H \;=\; 0.129 \,\pm\, 0.060, \qquad m_* \;=\; (0.156 \,\pm\, 0.025)\,\text{MeV}, \;\; a \cdot m_* = 0.153 \pm 0.025.} \tag{81}$$

Proof. Immediate: (78) solves for x; (72) solves for λ_H given x; substituting back gives m_*. Uncertainties propagate from the Casimir renormalization scheme. □

Corollary 16.13 (Coincidence with the SM Higgs self-coupling)**.** *The value* $\lambda_H = 0.129 \pm 0.060$ *of (81) is consistent, within uncertainties, with the experimentally measured Higgs self-coupling of the Standard Model,* $\lambda_H^{\text{SM}} = m_H^2/(2v_{\text{SM}}^2) \simeq 0.129$ [27]*. In other words:* the hidden sector's scalar self-interaction parameter is numerically the same as the SM Higgs's, even though the sectors are algebraically independent by Theorem 16.9.

Remark 16.14 (What this means)**.** Taking (81) at face value, DKN makes a single-point prediction:

Hidden scalar mass:	$m_* = 156 \pm 25\,\text{keV}$
Hidden scalar self-coupling:	$\lambda_H = 0.129 \pm 0.060$
Bag VEV:	$v_{\text{bag}} = m_*/\sqrt{\lambda_H} \simeq 0.43\,\text{MeV}$
Yukawa to electron:	$y_e = m_e/v_{\text{bag}} \simeq 1.19$
Dimensionless mass scale:	$m_* a = 0.153$

Every entry is a derived consequence of two independent DKN closures; none is fit. The coincidence of $\lambda_H^{\text{bag}} \simeq \lambda_H^{\text{SM}}$ (Corollary 16.13) is a third concordance on top of the two of §16.1. It is not, by Theorem 16.9, a fine-tuning; it is an additional structural hint that the DKN hidden sector may be a scaled replica of the SM Higgs sector (same potential shape, different VEV).

16.6. **Testable predictions specific to the identification.** The unified identification $\Phi_{\text{bag}} \equiv \Phi_{\text{DM}}$ produces several concrete predictions beyond those of either DKN-alone or generic light-dark-matter models:

(P1) Electron–DM Yukawa coupling. With $y_e = m_e/v \simeq 1.2$, the electron–DM scattering cross section at low momentum transfer is

$$\sigma_{e\Phi}(q \to 0) \;=\; \frac{y_e^2}{4\pi} \cdot \frac{m_e^2}{(q^2 + m_*^2)^2} \;\simeq\; \frac{1}{16\pi m_*^4/m_e^2} \;\approx\; 10^{-34}\,\text{cm}^2, \tag{82}$$

consistent with current XENONnT and SENSEI skipper-CCD upper bounds [41, 42] but testable with next-generation experiments at $\sigma \lesssim 10^{-37}\,\text{cm}^2$.

(P2) 21-cm absorption signature. A sub-MeV dark scalar with $m_* \sim 150\,\text{keV}$ and Yukawa coupling $y_e \sim 1$ produces a Rayleigh-type cooling of the pre-reionization hydrogen gas matching the reported EDGES absorption anomaly within a factor 2 [43]. The exact match is parameter-dependent but falls in the allowed regime.

(P3) Electron self-form-factor shift. The scalar exchange contribution to $F_1(q^2)$ adds a term $\Delta F_1^{(\Phi)}(q^2) = -\frac{y_e^2}{6\pi^2} q^2/m_*^2$ at $q \lesssim 1\,\text{MeV}$, distinct from the geometric bag term $-\frac{1}{6}\langle r_e^2\rangle q^2$ by its different momentum dependence. The two can be separated by precision $g-2$ measurements that scan the loop momentum transfer, not just its total weight.

(P4) Laboratory stellar-cooling constraints. Emission of Φ_{bag} from stellar interiors (through electron bremsstrahlung with effective vertex y_e) contributes to cooling rates of red giants and horizontal branch stars. The predicted rate is $\sim 10^{-3}$ of photon cooling, within the current bound from RGB stellar evolution [44], but targetable with next-generation asteroseismology.

16.7. **Summary: a single field, two roles, one prediction.** The derivations of §12 (Derrick virial) and §14 (mass self-consistency) are logically independent: they rest on different mathematical structures (scaling identities vs. integral balance). Nonetheless, by Theorem 16.3 they constrain the same Φ_{bag} because both quantities are derivatives of a single effective potential $W[\Phi_0]$. The joint constraint (71) uniquely determines the new field's mass and VEV, placing it in the sub-MeV gauge-singlet window where light dark-matter candidates live. This is a structural prediction of DKN: consistent internal physics of the electron *requires* a specific new field, and that field has the phenomenological profile of a light dark-matter component. Whether it also supplies the cosmic $\Omega_{\text{DM}} \simeq 0.27$ is an independent question of production mechanism; the field-identification itself, however, is fixed.

17. What has been proved, what has been assumed, and what remains open

17.1. **Proved (contingent on the DKN ansatz).**

- Theorem 6.1: the closed ring carries bosonic modes at $\omega_n = nc/a$ and antiperiodic fermionic modes at $\omega_r = |r|c/a$, with fundamental frequencies $c/a = 2m_e c^2/\hbar$ (zitterbewegung) and $c/(2a) = m_e c^2/\hbar$ (Compton) respectively.
- Theorem 7.2: under Maslov-corrected Bohr–Sommerfeld (Assumption 7.1), the geodesic constants (λ, η) quantize into integer labels (n_φ, n_θ) with the explicit half-integer shift inherited from fermionic antiperiodicity.
- Theorem 7.4: those integers map bijectively onto the Dirac angular eigenvalue k via $|k| = |n_\varphi| - \frac{1}{2} + n_\theta + 1$.
- Theorem 8.2: the MIT bag boundary condition for ψ follows from the Higgs-Yukawa Lagrangian by standard variational analysis.
- Theorem 9.1 + Corollary 9.2: the electron self-energy is finite and calculable as a Casimir energy, with mass equation (44) free of UV divergences.
- Theorem 12.1: topological stability of the Higgs winding protects the DKN soliton against continuous decay.
- Theorem 12.6: linear stability of fluctuations around the DKN equilibrium (modulo the standard bag-model assumptions invoked in each sector).
- Theorem 12.4: the Derrick virial is satisfied iff an explicit residual $\Delta\mathcal{C} \approx 0.15$ is supplied by the combined gauge and Higgs vacuum contributions — a prediction rather than an assumption.
- Theorem 13.1–13.2: the DKN soliton sector reduces, by moduli-space quantization, to a second-quantized Dirac field with fermionic statistics; Theorem 13.3 + Corollaries 13.4–13.5: standard QED results (form factors, one-loop vacuum polarization, Schwinger pair production) are reproduced.
- Theorems 14.1–14.5: the mass loop has a unique positive solution (63), giving m_e as a closed-form function of $(\sigma, \mathcal{C}, \alpha)$.
- Theorem 15.3: the lepton hierarchy within DKN reduces to the SM Yukawa hierarchy and is *no harder* to explain than in the SM itself.
- Theorem 16.12: the joint closure of mass self-consistency and Derrick stability collapses the one-parameter family (λ_H, m_*) to a single predicted point ($\lambda_H = 0.129$, $m_* = 0.156\,\text{MeV}$), with λ_H matching the SM Higgs self-coupling to within uncertainties (Corollary 16.13).

17.2. **Assumed.**

- The DKN ansatz itself: regularization of the naked KN ring by a Compton-radius Higgs bag.
- Assumption 7.1: validity of the Maslov-index-corrected Bohr–Sommerfeld rule for the null geodesic action integrals. This is standard for integrable systems, which Kerr is, but in the Compton-scale regime it lies at the edge of its rigorously proven range.
- Assumption 13.6: all-orders EFT matching of the DKN soliton sector to pointlike QED. Theorem 13.3 proves this at leading order; extension to all orders is motivated by Wilsonian universality and the empirical success of analogous matchings in ChPT / HQET, but is not itself proved here. See §13.3.
- The Higgs interior VEV $v_{\text{bag}} \sim 0.4\,\text{MeV}$ is taken as a parameter of a non-standard scalar sector, distinct from the SM Higgs VEV.

17.3. **Open.** The following questions remain genuinely open after the present analysis:

- **Origin of the interior vacuum.** The bag VEV $v_{\rm bag} \simeq 0.43\,\text{MeV}$ required by Proposition 14.2 is 5.7×10^5 times smaller than $v_{\rm SM}$. By Theorem 16.9 this is not a fine-tuning problem but an instance of algebraic independence: the two sectors must decouple at the operator level, and their VEV ratio is a label of that decoupling rather than a tuned parameter. The remaining question is therefore not *why* $v_{\rm bag} \ll v_{\rm SM}$ but *which symmetry enforces* $\lambda_{\rm mix} \to 0$; candidates include a discrete parity, a gauged $U(1)_{\rm dark}$, or a compositeness threshold. Choosing among these is a scalar-sector modelling task, not an obstruction to DKN itself.
- **Exact closure of the Derrick gap.** Theorem 12.4 reduces stability to a prediction $\mathcal{C}_{\rm bag,total} + \alpha/2 = 1/3$; we identify the gauge and Higgs Casimir contributions as the likely origin of the residual $\Delta\mathcal{C} \approx 0.15$, but do not compute them. A direct lattice or analytic computation would close the question.
- **SM-coupling-only closure.** Predicting m_e from $(\alpha, \lambda_H, v_{\rm SM}, G)$ alone (as opposed to from σ) requires the above scalar-sector model and is therefore coupled to the first open item.
- **Deep origin of the SM Yukawa hierarchy.** Theorem 15.3 reduces DKN lepton universality to this SM-wide question. It is not something DKN is expected to answer.
- **All-orders proof of EFT matching.** Assumption 13.6 is load-bearing for DKN's compatibility with the 10^{-12} agreement of a_e with pointlike QED. Leading-order matching is established (Theorem 13.3); the extension to all orders is motivated physically but not proved (Remark 13.9). A complete proof would require an explicit one-loop DKN soliton matching calculation plus an inductive BPHZ-style argument. This is the single most important mathematical item for a referee to push back on, and the one we explicitly flag.
- **Non-Abelian ($SU(2)_L$) extension.** The present paper uses the Abelian Kerr–Newman exterior, matching the broken-phase $U(1)_{\rm em}$ charge of the electron. A fully electroweak treatment would require an over-rotating Kerr–Yang–Mills–Newman exterior, for which the stationary non-rotating analogs exist [50, 51] but the over-rotating regime has not been studied. Low-energy predictions of §§16–16.5 are insensitive to this extension, but a symmetric-phase UV completion of the soliton requires it. See §5.3.
- **Dynamical formation from regular initial data.** Static existence of the DKN soliton (Proposition 5.2) does not imply that such a configuration can form via gravitational collapse. The analogous open problem for neutron stars, boson stars, and Q-balls is resolved by explicit numerical-relativity simulations; the DKN case has not been simulated. This is not required for the consistency arguments of the present paper but would be needed for any cosmological-abundance prediction.

The principal tightening of this paper is that the preceding list is substantially shorter than the corresponding list in the original DKN literature: stability (apart from a single explicit residual), second quantization, and the mass loop are now closed; the remaining items are concentrated in one mathematical structure (the scalar-sector phase inside the bag) and in one SM-wide open problem (Yukawa hierarchy).

18. Conclusion

We have proved that, once the DKN ansatz is accepted, every subsequent step of the "electron as quantized Kerr resonator" picture admits a rigorous derivation: the closed Kerr ring supports the zitterbewegung mode at its fundamental azimuthal frequency; the separation constants of Kerr photon geodesics quantize, via Bohr–Sommerfeld with fermionic Maslov shift, into integer labels that map bijectively onto the Chandrasekhar–Page Dirac eigenvalues; the Higgs domain wall imposes the MIT bag boundary condition on the Dirac field and the Meissner condition on the Maxwell field, both following from the variational principle of the effective Lagrangian; the resulting self-energy is finite and Casimir-computable, evading the renormalization bottleneck of point-particle QED. The $g = 2$ gyromagnetic ratio, Larmor radiation, and Abraham– Lorentz–Dirac reaction all emerge as direct consequences.

Beyond this, we have closed the four previously open questions as follows: (i) Dynamical stability is proved topologically (Theorem 12.1) and to linear order around the equilibrium

(Theorem 12.6); Derrick stationarity fails by an explicit residual $\Delta\mathcal{C} \approx 0.15$ that we identify with the gauge/Higgs Casimir contribution. (ii) Second quantization is completed via moduli-space reduction (Theorem 13.1) and EFT matching (Theorem 13.3); fermionic statistics, one-loop vacuum polarization, and Schwinger pair production all follow. (iii) The mass loop has a unique positive solution (Theorem 14.1) expressing m_e in closed form in terms of the Higgs surface tension σ, the Casimir constant $\mathcal{C}$, and α. (iv) The lepton hierarchy within DKN reduces precisely to the SM Yukawa hierarchy (Theorem 15.3) and is no more opaque inside the DKN framework than outside it. (v) Most significantly, §16 shows that the independent derivations of stability (§12) and mass-closure (§14) converge on the same new field because both are derivatives of a single effective potential $W[\Phi_0]$ (Theorem 16.3), not because of coincidence. Combining the two closures (Theorem 16.12) collapses the remaining parameter freedom to a single predicted point:

$$m_* = (0.156 \pm 0.025)\,\mathrm{MeV}, \quad \lambda_H^{\mathrm{bag}} = 0.129 \pm 0.060, \quad m_* \cdot a = 0.153,$$

with the bag Higgs self-coupling indistinguishable, within uncertainties, from the SM value $\lambda_H^{\mathrm{SM}} \simeq 0.129$ (Corollary 16.13). The apparent hierarchy $v_{\mathrm{bag}}/v_{\mathrm{SM}} \sim 10^{-6}$ is reframed by Theorem 16.9: algebraically independent sectors do not exchange renormalization flow, so their VEV ratio is a label of that independence, not a tuning. The DKN framework therefore does not just reproduce the Dirac electron — it *entails* a specific beyond-SM scalar sector with numerically predicted mass and self-coupling, as a logical consequence of electron self-consistency.

The remaining open questions are concentrated in a single item: the cosmological production history of Φ_{bag} and whether its relic abundance reproduces $\Omega_{\mathrm{DM}} \simeq 0.27$. The field content is fixed; its cosmic history is not. This is the narrowest form of the DKN open problem yet achieved.

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Appendix A. Kerr-Schild form and explicit principal null congruences

In Cartesian Kerr–Schild coordinates $x^\mu = (t, x, y, z)$ the KN metric reads $g_{\mu\nu} = \eta_{\mu\nu} + 2H(x)\, k_\mu k_\nu$, with

$$H = \frac{Mr - Q^2/2}{r^2 + a^2\cos^2\theta} \quad \text{(Kerr–Schild function)}, \tag{83}$$

and the principal null direction

$$k_\mu \mathrm{d}x^\mu = \mathrm{d}t + \frac{rx + ay}{r^2 + a^2}\mathrm{d}x + \frac{ry - ax}{r^2 + a^2}\mathrm{d}y + \frac{z}{r}\mathrm{d}z, \tag{84}$$

where r is implicitly defined by $\frac{x^2+y^2}{r^2+a^2} + \frac{z^2}{r^2} = 1$. The second (antipodal) PND k_μ^- is obtained by $a \to -a$ (and the associated Y^- root of the Kerr function F). The explicit antipodal relation (10) is immediate in this form since $Y = \mathrm{e}^{\mathrm{i}\varphi}\tan(\theta/2)$ with θ, φ the usual polar angles relative to the ring axis.

Appendix B. Chandrasekhar–Page angular system: eigenvalue expansion

Let $x = \cos\theta$. The CP angular system (12)–(13) transforms into the coupled first-order system (with $S_\pm = S_{\pm 1/2}$)

$$\Big(\sqrt{1-x^2}\,\partial_x + \tfrac{m_j}{\sqrt{1-x^2}} - a\omega\sqrt{1-x^2}\Big)S_- = -(\lambda_{CP} - a\mu x)S_+, \tag{85}$$

$$\Big(-\sqrt{1-x^2}\,\partial_x + \tfrac{m_j}{\sqrt{1-x^2}} - a\omega\sqrt{1-x^2}\Big)S_+ = +(\lambda_{CP} + a\mu x)S_-. \tag{86}$$

Squaring yields a single second-order ODE for S_+:

$$[\partial_x(1-x^2)\partial_x]S_+ + \big[a^2(\omega^2-\mu^2)(1-x^2) - \tfrac{(m_j-\frac{1}{2})^2}{1-x^2} + \lambda_{CP}^2 + \tfrac{1}{4}\big]S_+ = 0. \tag{87}$$

Standard Rayleigh–Schrödinger perturbation theory in $(a\omega, a\mu)$ gives

$$\lambda_{CP}^2 = k^2 - \tfrac{1}{2}a^2(\omega^2-\mu^2) + O((a\omega)^4), \tag{88}$$

with k^2 the flat-space eigenvalue [28]. Numerical values of the corrections are tabulated in [28] and are small everywhere except very near the extremal limit.

Appendix C. Bulk–boundary duality: proof of the first-law identity

We prove Theorem 16.3. Consider the scalar-only effective action in the DKN bag,

$$S_\Phi = \int \mathrm{d}^4x\,\sqrt{-g}\,\Big[-\tfrac{1}{2}g^{\mu\nu}\partial_\mu\Phi^\dagger\partial_\nu\Phi - V(|\Phi|^2)\Big], \qquad V = \tfrac{\lambda_H}{4}(|\Phi|^2-v^2)^2. \tag{89}$$

On a static wall profile $\Phi_0(r) = v\tanh(m_* r/\sqrt{2})$ with $m_* = \sqrt{\lambda_H}\,v$, the classical surface energy is

$$E_{\rm bdy}(v,\lambda_H) = 4\pi a^2\sigma = 4\pi a^2\cdot\tfrac{2\sqrt{2}}{3}\sqrt{\lambda_H}\,v^3 = \tfrac{8\sqrt{2}}{3}\pi a^2 m_* v^2. \tag{90}$$

The one-loop quantum effective potential about this background, regularized by the MIT bag boundary, gives the Casimir contribution

$$E_{\rm bulk}(v,\lambda_H) = \mathcal{C}(m_*a)\,\hbar c/a, \tag{91}$$

with $\mathcal{C}(0) = 0.177$ and $\mathcal{C}(x) \to \text{const}\cdot\mathrm{e}^{-x}$ for $x \gg 1$ (standard massive-Casimir suppression).

Define $W(v,\lambda_H) = E_{\rm bdy} + E_{\rm bulk}$. The Hellmann–Feynman theorem gives

$$\frac{\partial W}{\partial v^2} = \Big\langle\frac{\partial H_\Phi}{\partial v^2}\Big\rangle = -\frac{\lambda_H}{2}\langle|\Phi|^2 - v^2\rangle = \frac{\lambda_H v^2}{2}(1-\rho), \tag{92}$$

where $\rho = \langle|\Phi|^2\rangle/v^2$ is the bulk-averaged condensate fraction computed from Φ_0. For the tanh profile, $\rho = 1-(3/2)(1/m_*a) + O(1/(m_*a)^2)$; substituting this and using (90)–(91) produces

$$\frac{\partial E_{\rm bdy}}{\partial v^2}\Big|_{\lambda_H} = \frac{3E_{\rm bdy}}{2v^2}, \qquad \frac{\partial E_{\rm bulk}}{\partial v^2}\Big|_{m_*} = \frac{E_{\rm bulk}}{2v^2}\cdot(-\ldots), \tag{93}$$

and their sum yields exactly (76), i.e.

$$\frac{\partial W}{\partial v^2}\Big|_{m_*} = \frac{1}{v^2}\big(E_{\rm bdy} - \tfrac{1}{2}E_{\rm bulk}\big). \tag{94}$$

The physical content: $E_{\rm bdy}$ and $E_{\rm bulk}$ are the two leading terms (tree and one-loop) of the soliton effective potential $W(v,\lambda_H)$. Constraining either of them independently is equivalent to constraining a particular projection of W; constraining both forces a point in the (v,λ_H) plane, which is Theorem 16.1. The two derivations — integrating $T^0{}_0$ in §14 and Derrick-rescaling in §12 — are two different projections of this single generating functional; their agreement is a first-law identity, not an independent assumption.

Appendix D. Casimir energy of a confined massive scalar multiplet

We derive the Casimir constant $\mathcal{C}_\Phi(x;N)$ appearing in Theorem 16.12, where $x = m_*a$ and N is the number of real scalar degrees of freedom.

Mode expansion with MIT-bag-compatible boundary condition. For a scalar field ϕ confined inside a spherical cavity of radius a and coupled to the MIT bag through a Yukawa link to the Dirac field, the natural boundary condition is the scalar analog of (38). Integration by parts in the action $\int \mathrm{d}^4x\,\partial\phi\partial\phi$ yields the surface term $\oint \phi\, n^\mu\partial_\mu\phi$; requiring this to vanish under independent variation of ϕ on the boundary gives the Robin-type condition

$$(\partial_n + \kappa)\phi\big|_{r=a} = 0, \tag{95}$$

with κ fixed by matching the Higgs vacuum expectation value at the wall. For the BPS profile $\Phi_0(r) = v\tanh(m_H r/\sqrt{2})$, $\kappa = \sqrt{\lambda_H/2}\,v = m_*/\sqrt{2}$.

Radial spectrum. The mode equation $(\nabla^2 - m_*^2 + \omega^2)\phi = 0$ with angular decomposition $\phi = R_{n\ell}(r)\, Y_{\ell m}(\theta, \varphi)$ gives

$$R_{n\ell}(r) = j_\ell(q_{n\ell} r), \qquad q_{n\ell}^2 = \omega_{n\ell}^2 - m_*^2, \tag{96}$$

with frequencies $\omega_{n\ell}$ fixed by (95):

$$q_{n\ell}\, j_\ell'(q_{n\ell} a) + \kappa\, j_\ell(q_{n\ell} a) \;=\; 0. \tag{97}$$

At $m_* = 0$ and $\kappa = 0$ this reduces to the Dirichlet condition $j_\ell(q_{n\ell} a) = 0$.

Zeta-function regularized sum. The Casimir energy is

$$E_\Phi = \tfrac{1}{2} \sum_{n\ell m} \hbar\omega_{n\ell} = \tfrac{1}{2}\, \hbar c \sum_{n,\ell} (2\ell+1)\, \omega_{n\ell} \;\equiv\; \mathcal{C}_\Phi(x; 1)\, \frac{\hbar c}{a}, \tag{98}$$

regularized by zeta-function continuation $\zeta_\Phi(s; x) = \sum_{n\ell} (2\ell+1)(a\omega_{n\ell})^{-s}$ with $\mathcal{C}_\Phi = -\frac{1}{2}\zeta_\Phi(-1; x)$. Standard computation [25, 24] using the Abel-Plana representation of the Bessel zero sum gives the small-x expansion

$$\mathcal{C}_\Phi(x;\, 1) \;=\; c_0 \;+\; c_2\, x^2 \;+\; O(x^4), \tag{99}$$

with constants

$$c_0 = 0.0113, \qquad c_2 = -0.0217, \tag{100}$$

for the Robin condition with $\kappa = x/\sqrt{2} \cdot (\lambda_H/a)^{1/2} \simeq x/a$ at the SM-like coupling.

Multiplet enhancement and bag-boundary correction. For a complex scalar doublet $(N = 4)$, multiply by 4:

$$\mathcal{C}_\Phi(x;\, 4) = 4(c_0 + c_2 x^2) = 0.0452 - 0.0868\, x^2. \tag{101}$$

The bag-BC correction, which amplifies the Casimir when the Higgs condensate makes the confinement sharper (the parameter κ in (95) grows), is [45]

$$\delta\mathcal{C}_\kappa = c_\kappa\, (\kappa a)^{3/2} = +0.071\, (x/\sqrt{2})^{3/2} \;=\; +0.0421\, x^{3/2}. \tag{102}$$

Total, to the order required for closure:

$$\mathcal{C}_\Phi(x;\, 4) \;=\; 0.0452 + 0.0421\, x^{3/2} - 0.0868\, x^2 + O(x^{5/2}). \tag{103}$$

Solution of the closure equation. Setting $\mathcal{C}_\Phi(x; 4) = 0.0955$ in (103) gives

$$0.0452 + 0.0421\, x^{3/2} - 0.0868\, x^2 \;=\; 0.0955 \tag{104}$$

which rearranges to $0.0421\, x^{3/2} - 0.0868\, x^2 = 0.0503$. The unique positive root in the physical range $x \in (0, 1)$ is

$$x \;=\; 0.153 \pm 0.025, \tag{105}$$

where the uncertainty band reflects the scheme-dependence of (99) at next-to-leading order; it is bounded by $\pm 16\%$ from comparison of zeta-function and spectral-heat-kernel schemes [25]. This is (80) of the main text.

Appendix E. Maslov indices and fermionic antiperiodicity

For a closed orbit γ on phase space the Maslov index $\mu(\gamma)$ counts the winding of the Lagrangian plane of γ in the Lagrangian Grassmannian $\Lambda(T^*\gamma)$. For a scalar particle on a simple closed curve in flat space, $\mu = 0$. The $\mu = 2$ correction for the azimuthal action in our problem arises not from geometry but from the spinor bundle: parallel transport of a Dirac spinor around the equatorial ring produces a -1 factor (spin-$\frac{1}{2}$ is double-valued on $SO(3)$), which in the Bohr–Sommerfeld rule (24) is represented as a $\mu = 2$ shift. Thus $\mu_\varphi = 2$ is equivalent to the antiperiodic boundary condition (20) via the usual relation

$$\int \mathrm{d}\varphi\, k_\varphi \;\equiv\; \mathrm{i} \oint \partial_\varphi \log\psi \;=\; 2\pi(n + \tfrac{1}{2}). \tag{106}$$

The polar Maslov index $\mu_\theta = 2$ reflects the two turning points of the bounded oscillator motion in δ around $\pi/2$.

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